BME302 Unit I – Fluid Properties & Bernoulli’s Equation

Lecture Notes · Fluid Mechanics and Fluid Machines

Dr. Abhishek Pandey, Department of Mechanical Engineering, G. L. Bajaj Institute of Technology & Management

How to use these notes. Every topic, worked example and question carries one tag. CORE must-know syllabus material at AKTU exam level: master these first. HOTS higher-order thinking: analyse, compare, justify or design (Bloom K4 to K6). ADVANCED beyond the exam level for most students: shaded orange, optional reading.

Unit overview

Unit I builds the vocabulary of the subject: what a fluid is, the properties that decide how it behaves, and the one energy equation (Bernoulli’s) behind the three common flow meters. It maps to **CO1: Understand the application of mass and momentum conservation laws for fluid flows (K2)** and is planned for 8 lectures; notation follows R.K. Bansal and Som, Biswas & Chakraborty.

Lecture	Topic	Key outcome
L1	Definition of fluid, continuum, Newton’s law of viscosity, types of fluids	Tell a fluid from a solid; classify fluids
L2	Units and dimensions; density, specific weight, specific volume, specific gravity	Convert units; compute mass properties
L3	Viscosity: dynamic and kinematic, effect of temperature	Shear stress, torque and power in viscous problems
L4	Compressibility and bulk modulus; surface tension	Volume change under pressure; pressure inside drops
L5	Capillarity	Capillary rise and depression
L6	Incompressible flow: Euler’s equation and Bernoulli’s equation	Energy heads along a streamline
L7	Venturimeter and orifice meter	Discharge from a manometer reading
L8	Pitot tube; revision	Point velocity from stagnation pressure

Common notation: ρ = mass density, w = specific weight, v = specific volume, S = specific gravity, μ = dynamic viscosity, ν = kinematic viscosity, K = bulk modulus, σ = surface tension, θ = contact angle, p = pressure, V = velocity, z = datum head, Q = discharge, C_d =

coefficient of discharge, C_v = coefficient of velocity. Subscripts $_1$ and $_2$ mark the inlet and throat (or upstream and downstream) sections.

0. Before you start: basic ideas you need CORE

This section collects the basic ideas that the whole unit uses again and again. Read it first if the problems feel hard to start. Every later section assumes you are at ease with units, pressure and the difference between mass and weight.

> **IN SIMPLE WORDS**
> Fluid mechanics is mostly careful bookkeeping of units. A pressure can be written as a force on an area, or as the height of a liquid column that would create it. Mass is how much matter there is, in kg. Weight is the pull of gravity on that mass, in N. If you convert every number to SI units first and keep mass and weight apart, most numerical problems in this unit become simple substitution.

0.1 Quantities, units and conversions CORE

Most wrong answers come from units, not from physics. Convert every value to SI units (m, kg, s, N, Pa) before you put it into a formula.

Symbol	Meaning	SI unit	Typical value
ρ	Mass density	$\mathrm{kg/m^3}$	Water 1000; mercury $13\,600$; air 1.2
$w = \rho g$	Specific weight	$\mathrm{N/m^3}$	Water 9810
S	Specific gravity	No unit	Oils 0.8 to 0.95; mercury 13.6
μ	Dynamic viscosity	$\mathrm{Pa{\cdot}s}$ (same as $\mathrm{N\cdot s/m^2}$)	Water 1.0×10^{-3} at $20^{\circ}\mathrm{C}$
$\nu = \mu/\rho$	Kinematic viscosity	$\mathrm{m^2/s}$	Water 1.0×10^{-6}
p	Pressure	$\mathrm{Pa} = \mathrm{N/m^2}$	Atmosphere about $101.3\,\mathrm{kPa}$
K	Bulk modulus	$\mathrm{N/m^2}$	Water about 2.1×10^9
σ	Surface tension	$\mathrm{N/m}$	Water 0.0725
V	Velocity	$\mathrm{m/s}$	1 to $3\,\mathrm{m/s}$ in water pipes
A, a	Cross-sectional area	$\mathrm{m^2}$	$0.0707\,\mathrm{m^2}$ for $D = 0.3\,\mathrm{m}$
Q	Discharge	$\mathrm{m^3/s}$	0.001 to $1\,\mathrm{m^3/s}$
g	Acceleration due to gravity	$\mathrm{m/s^2}$	9.81

Conversions that students often get wrong.

Given in	Convert to	Rule	Example
mm	m	Divide by 1000	$80\ \text{mm} = 0.08\ \text{m}$
cm/s	m/s	Divide by 100	$60\ \text{cm/s} = 0.6\ \text{m/s}$
cm^2	m^2	Divide by 10^4	$25\ \text{cm}^2 = 0.0025\ \text{m}^2$
L/s	m^3/s	Divide by 1000	$15\ \text{L/s} = 0.015\ \text{m}^3/\text{s}$
N/cm^2	N/m^2	Multiply by 10^4	$70\ \text{N/cm}^2 = 7 \times 10^5\ \text{N/m}^2$
bar	N/m^2	Multiply by 10^5	$2\ \text{bar} = 2 \times 10^5\ \text{N/m}^2$
kPa	m of water	Change to Pa, then $h = p/(\rho g)$	$100\ \text{kPa} \to 100\,000/(1000 \times 9.81) = 10.19\ \text{m}$
poise (P)	Pa·s	$1\ \text{P} = 0.1\ \text{Pa}\cdot\text{s}$	$8\ \text{P} = 0.8\ \text{Pa}\cdot\text{s}$
centipoise (cP)	Pa·s	$1\ \text{cP} = 10^{-3}\ \text{Pa}\cdot\text{s}$	$12\ \text{cP} = 0.012\ \text{Pa}\cdot\text{s}$
stokes (St)	m^2/s	$1\ \text{St} = 10^{-4}\ \text{m}^2/\text{s}$	$3\ \text{St} = 3 \times 10^{-4}\ \text{m}^2/\text{s}$
rpm	rad/s	$\omega = 2\pi N/60$	$1500\ \text{rpm} \to 157.1\ \text{rad/s}$
km/h	m/s	Divide by 3.6	$72\ \text{km/h} = 20\ \text{m/s}$

Example 0.1 CORE Convert to SI units: (a) 12 cP, (b) 3 stokes, (c) 45 cm², (d) 600 rpm.

1. $12\ \text{cP} = 12 \times 10^{-3} = 0.012\ \text{Pa}\cdot\text{s}$
2. $3\ \text{St} = 3 \times 10^{-4} = 3 \times 10^{-4}\ \text{m}^2/\text{s}$
3. $45\ \text{cm}^2 = \dfrac{45}{10^4} = 0.0045\ \text{m}^2$
4. $\omega = \dfrac{2\pi N}{60} = \dfrac{2\pi \times 600}{60} = 62.83\ \text{rad/s}$

0.2 Pressure and its measurement CORE

Pressure is force per unit area, $p = F/A$, in pascals ($1\ \text{Pa} = 1\ \text{N/m}^2$). Two scales are in use:

- **Absolute pressure** is measured from a perfect vacuum.
- **Gauge pressure** is measured from the local atmosphere. A gauge on an open tank reads zero.

$$p_{\text{abs}} = p_{\text{atm}} + p_{\text{gauge}}$$

A negative gauge pressure is a **vacuum**. Use the same scale (gauge or absolute) at every point in one problem.

At a depth h below the free surface of a liquid at rest, the gauge pressure is:

$$p = \rho g h$$

Derivation from scratch: pressure at a depth, $p = \rho g h$

Aim. Show that the gauge pressure at depth h in a liquid at rest is $\rho g h$.

Assumptions.

1. The liquid is at rest and its density ρ is constant.
2. The pressure on the free surface is p_0 (the atmosphere).
3. The liquid column is a vertical cylinder of cross-sectional area A and height h.

Step 1. Weight of the column. The volume is Ah, the mass is $\rho A h$, and the weight is:

$$W = \rho A h\, g$$

Step 2. Forces on the column. The air pushes down on the top face with force $p_0 A$. The weight W acts downward. The liquid below pushes up on the bottom face with force pA, where p is the pressure at depth h. The sides carry only horizontal forces, which cancel.

Step 3. Equilibrium. The column is at rest, so the upward force equals the downward forces.

$$pA = p_0 A + \rho g A h$$

Step 4. Cancel A.

$$p = p_0 + \rho g h$$

Result. Measured from the atmosphere ($p_0 = 0$ gauge):

$$p = \rho g h \quad \Rightarrow \quad h = \frac{p}{\rho g}$$

What it tells us. Pressure grows in direct proportion to depth and to density. A pressure can be quoted as a height of liquid. For the same pressure, mercury ($\rho = 13\,600$) needs only $1/13.6$ of the height of water.

This idea is the basis of the **manometer**, which measures a pressure difference from the height of a liquid column. Two points at the same level in the same connected liquid at rest have the same pressure. Standard atmosphere:
$1\ \text{atm} = 101.325\ \text{kPa} = 760\ \text{mm of mercury} = 10.33\ \text{m of water}$.

Example 0.2 CORE A gauge at the bottom of an open tank of water, 4 m deep, is read. Find the gauge and the absolute pressure ($p_{\text{atm}} = 101.3\ \text{kPa}$).

1. $p_{\text{gauge}} = \rho g h = 1000 \times 9.81 \times 4 = \ 39\,240\ \text{Pa}\ \ (\ 39.24\ \text{kPa}\ )$
2. $p_{\text{abs}} = 101.3 + 39.24 = \ 140.5\ \text{kPa}$

0.3 Density, specific gravity, and the units of force and mass CORE

Do not mix **mass** and **weight**. Mass m is in kg. Weight is a force, $W = mg$, in newtons. A body of 1 kg has a weight of 9.81 N. The older unit $1\ \text{kgf} = 9.81\ \text{N}$ is also a force. The small letter g means gravity in formulas and also “gram” as a unit. A gram is 10^{-3} kg.

Quantity	Rule	Example
Density	$1\ \mathrm{g/cm^3} = 1000\ \mathrm{kg/m^3}$	$0.85\ \mathrm{g/cm^3} = 850\ \mathrm{kg/m^3}$
Specific gravity	$S = \rho/\rho_{\text{water}}$, with $\rho_{\text{water}} = 1000\ \mathrm{kg/m^3}$	Mercury: $S = 13.6$, $\rho = 13\,600\ \mathrm{kg/m^3}$
Weight	$W = mg$	$27.2\ \mathrm{kg} \to 266.8\ \mathrm{N}$
Volume	$1\ \mathrm{L} = 10^{-3}\ \mathrm{m^3}$	$2\ \mathrm{L} = 0.002\ \mathrm{m^3}$

Example 0.3 CORE Find the mass and the weight of 2 litres of mercury ($S = 13.6$).

1. $\rho = 13.6 \times 1000 = 13\,600\ \mathrm{kg/m^3}$; $V = 2\ \mathrm{L} = 0.002\ \mathrm{m^3}$
2. $m = \rho V = 13\,600 \times 0.002 = \ 27.2\ \mathrm{kg}$
3. $W = mg = 27.2 \times 9.81 = \ 266.8\ \mathrm{N}$

Common mistakes and exam tips.

- Putting mm, cm or litres straight into a formula. Convert to m, m² or m³ first.
- Mixing up poise and $\mathrm{Pa{\cdot}s}$: $1\ \mathrm{P} = 0.1\ \mathrm{Pa \cdot s}$, not 1. And $1\ \mathrm{St} = 10^{-4}\ \mathrm{m^2/s}$, not 10^{-2}.
- Mixing gauge and absolute pressure. Add p_{atm} only when the question asks for absolute pressure or uses a gas law.
- Forgetting g: a mass of 27.2 kg is not a weight of 27.2 N.
- Taking the density of water for every liquid. For oil of specific gravity S use $\rho = 1000\,S$.

Check your understanding.

1. CORE Convert 5 L/s to $\mathrm{m^3/s}$. *(It is $0.005\ \mathrm{m^3/s}$.)*
2. CORE A gauge reads 50 kPa and the atmosphere is 100 kPa. What is the absolute pressure? *(It is $150\ \mathrm{kPa}$.)*
3. HOTS Why does a mercury column balance the same pressure with a much shorter height than water? *(From $h = p/(\rho g)$: the density is 13.6 times larger, so h is 13.6 times smaller.)*

1. Fluid, continuum and Newton’s law of viscosity (L1) CORE

A **fluid** is a substance that deforms continuously under a shear stress, however small. A solid resists shear by a fixed deformation; a fluid at rest can carry no shear stress at all. Liquids and gases are both fluids: a liquid keeps its volume and forms a free surface, a gas fills its container.

IN SIMPLE WORDS

A fluid is anything that keeps flowing when you push it sideways. A brick resists a sideways push and stops moving. Water and air do not. Think of a pack of cards: a sideways push makes the layers slide over each other. The sliding resistance between layers is called viscosity, and it is the main idea of this section.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

1.1 Fluid as a continuum ADVANCED

Fluid mechanics treats a fluid as a **continuum**: properties such as density, pressure and velocity are taken as continuous functions of position, ignoring the gaps between molecules. The assumption holds when the mean free path λ of the molecules is tiny compared with the length scale L of the problem, measured by the Knudsen number:

$$Kn = \frac{\lambda}{L}, \qquad Kn < 0.01 \ (\text{continuum})$$

It fails in very rarefied gases (upper atmosphere, vacuum systems), where free molecular flow analysis is needed.

1.2 Newton's law of viscosity CORE

IN SIMPLE WORDS

Picture two layers of honey. If you drag the top layer, the layer below is dragged along, but more slowly. The shear stress τ is the drag force per unit area between the layers. The velocity gradient du/dy tells how fast the speed changes from one layer to the next. The viscosity μ is the constant that links them: thicker honey has a larger μ than water.

Consider fluid between a fixed plate and a plate moving at velocity U. The layer touching each plate moves with it (no slip), so the velocity rises from zero to U across the gap. The shear stress between layers is proportional to the rate of shear strain, which equals the velocity gradient:

$$\tau = \mu \frac{du}{dy}$$

The constant μ is the **dynamic viscosity**. For a thin gap of thickness t with a linear profile, $du/dy = U/t$, so $\tau = \mu U/t$.

1.3 Types of fluids CORE

IN SIMPLE WORDS

Fluids are sorted by how τ and du/dy are related. For a Newtonian fluid (water, air) the relation is a straight line. For others it is not. Ketchup gets thinner when you shake it. Wet sand gets stiffer when you hit it. Toothpaste does not flow until you squeeze hard enough.

Fluid	Behaviour	Example
Ideal	No viscosity, incompressible (imaginary)	Used in theory
Real	Has viscosity	All actual fluids
Newtonian	τ proportional to du/dy; μ constant at a given temperature	Water, air, light oils
Non-Newtonian: pseudoplastic (shear-thinning)	Apparent viscosity falls as du/dy rises	Blood, paints, polymer solutions
Non-Newtonian: dilatant (shear-thickening)	Apparent viscosity rises as du/dy rises	Starch–water mix, wet sand
Bingham plastic	No flow until a yield stress τ_0 is exceeded, then linear	Toothpaste, sewage sludge, drilling mud
Ideal plastic	Yield stress, then constant slope	Idealisation of Bingham behaviour

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

Non-Newtonian behaviour is commonly written as the power law $\tau = A\left(du/dy\right)^{n} + B$: $n < 1$ for pseudoplastic, $n > 1$ for dilatant, $B = \tau_0$ for Bingham plastic. The study of this behaviour is called **rheology**.

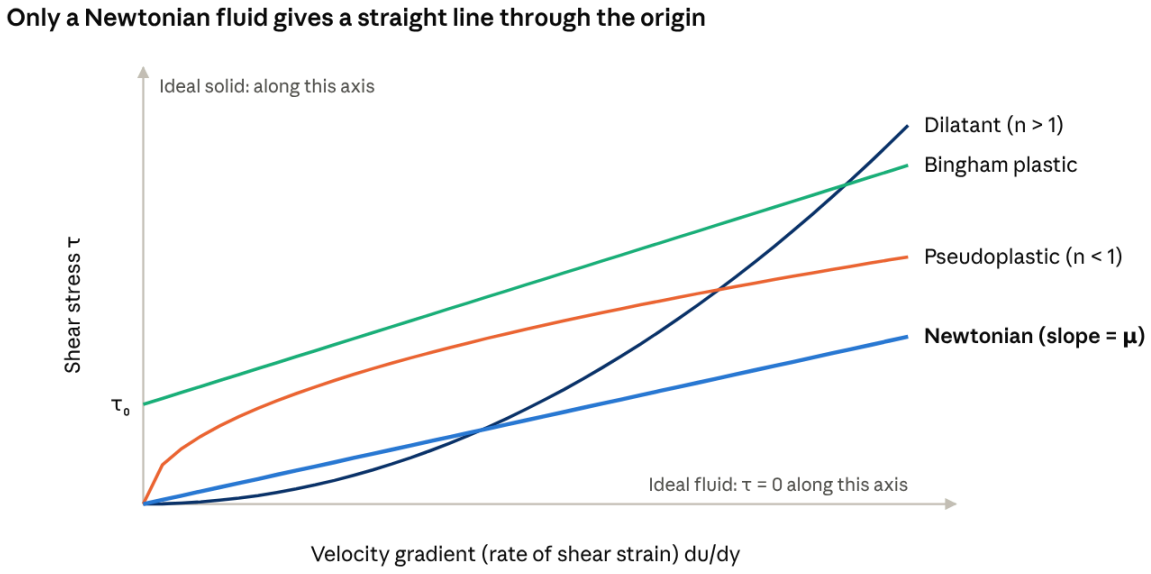


Figure 1. Rheological diagram: shear stress against velocity gradient

The slope of each curve at a point is the apparent viscosity there: constant for a Newtonian fluid, falling for a pseudoplastic, rising for a dilatant.

Common mistakes and exam tips.

- Saying a fluid at rest carries shear stress. It cannot. Shear stress appears only when layers move relative to each other.
- Using the plate speed U instead of the gradient du/dy. For a thin film the gradient is U/t, so keep the gap t in metres.
- Writing that every fluid is Newtonian. Blood, paint and toothpaste are not, and μ is not constant for them.
- Confusing ideal and real fluids. An ideal fluid has no viscosity and is only a model. All real fluids are viscous.

Check your understanding.

1. CORE Can a fluid at rest carry a shear stress? *(No. A fluid at rest carries no shear stress.)*
2. CORE What is the fluid velocity at the surface of a fixed plate? *(Zero, because of the no-slip condition.)*
3. HOTS Why must toothpaste be squeezed before it flows? *(It is a Bingham plastic: it needs a stress above its yield stress τ_0.)*

2. Units and dimensions (L2) CORE

Every fluid quantity can be written in terms of three fundamental dimensions, mass M, length L and time T (plus temperature θ for heat problems). SI units are used throughout: kg, m, s, with force in newtons ($1\,\mathrm{N} = 1\,\mathrm{kg}\cdot\mathrm{m/s^2}$).

IN SIMPLE WORDS

Every physical quantity is built from a few basic ones: mass, length and time. Writing a quantity in these terms gives its dimensions. Writing it with a unit gives a number you can measure. A formula is only correct if all its terms have the same dimensions. You cannot add metres to seconds. That simple rule lets you catch many mistakes before you calculate.

Quantity	Symbol	SI unit	Dimensions
Mass density	ρ	kg/m³	$\mathrm{ML^{-3}}$
Specific weight	w	N/m³	$\mathrm{ML^{-2}T^{-2}}$
Specific volume	v	m³/kg	$\mathrm{M^{-1}L^{3}}$
Pressure, shear stress, bulk modulus	p, τ, K	N/m² (Pa)	$\mathrm{ML^{-1}T^{-2}}$
Dynamic viscosity	μ	N·s/m² (Pa·s)	$\mathrm{ML^{-1}T^{-1}}$
Kinematic viscosity	ν	m²/s	$\mathrm{L^{2}T^{-1}}$
Surface tension	σ	N/m	$\mathrm{MT^{-2}}$
Velocity	V	m/s	$\mathrm{LT^{-1}}$
Discharge	Q	m³/s	$\mathrm{L^{3}T^{-1}}$
Power	P	W (N·m/s)	$\mathrm{ML^{2}T^{-3}}$

Useful conversions:

- $1\ \text{poise} = 1\ \text{dyne}\cdot\text{s/cm}^2 = 0.1\ \text{N}\cdot\text{s/m}^2$; $1\ \text{centipoise} = 10^{-3}\ \text{N}\cdot\text{s/m}^2$ (water at 20 °C is about 1 cP)
- $1\ \text{stoke} = 1\ \text{cm}^2/\text{s} = 10^{-4}\ \text{m}^2/\text{s}$
- $1\ \text{bar} = 10^{5}\ \text{N/m}^2$; $1\ \text{N/cm}^2 = 10^{4}\ \text{N/m}^2$; $1\ \text{atm} = 101.325\ \text{kN/m}^2 = 10.33\ \text{m of water}$

Every correct physical equation is **dimensionally homogeneous**: each term carries the same dimensions. Bernoulli's equation, for example, has every term in metres of fluid (energy per unit weight). This check catches algebra errors quickly and is the basis of dimensional analysis in later units.

Derivation from scratch: dimensional homogeneity of Bernoulli's equation, and the dimensions of μ

Aim. Show that every term of Bernoulli's equation has the dimension of length, and derive the dimensions of μ and ν used in the table above.

Assumptions.

1. The base dimensions are mass M, length L and time T.
2. Force follows Newton's second law, $F = ma$, so $[\,F\,] = \mathrm{MLT^{-2}}$. Pure numbers (such as 2 in $2g$) have no dimensions.

Step 1. Dimensions of pressure. Pressure is force per area.

$$[\,p\,] = \frac{\mathrm{MLT^{-2}}}{\mathrm{L^{2}}} = \mathrm{ML^{-1}T^{-2}}$$

Step 2. Pressure head. Use $[\,\rho\,] = \mathrm{ML^{-3}}$ and $[\,g\,] = \mathrm{LT^{-2}}$. First the product ρg:

$$[\,\rho g\,] = \mathrm{ML^{-3}} \times \mathrm{LT^{-2}} = \mathrm{ML^{-2}T^{-2}}$$

$$\left[\frac{p}{\rho g}\right] = \frac{\mathrm{ML^{-1}T^{-2}}}{\mathrm{ML^{-2}T^{-2}}} = \mathrm{L}$$

Step 3. Velocity head. Velocity has dimensions LT^{-1}.

$$\left[\frac{V^2}{2g}\right] = \frac{\left(\mathrm{LT}^{-1}\right)^2}{\mathrm{LT}^{-2}} = \frac{\mathrm{L}^2\mathrm{T}^{-2}}{\mathrm{LT}^{-2}} = \mathrm{L}$$

Step 4. Datum head. The height z is a length, so $[\,z\,] = \mathrm{L}$. All three terms of Bernoulli's equation are lengths, so the equation is dimensionally homogeneous.

Step 5. Dimensions of μ. Newton's law gives $\mu = \tau/\left(\,du/dy\,\right)$. The shear stress has the dimensions of pressure. The velocity gradient is:

$$\left[\frac{du}{dy}\right] = \frac{\mathrm{LT}^{-1}}{\mathrm{L}} = \mathrm{T}^{-1}$$

$$[\,\mu\,] = \frac{\mathrm{ML}^{-1}\mathrm{T}^{-2}}{\mathrm{T}^{-1}} = \mathrm{ML}^{-1}\mathrm{T}^{-1}, \qquad [\,\nu\,] = \frac{[\,\mu\,]}{[\,\rho\,]} = \frac{\mathrm{ML}^{-1}\mathrm{T}^{-1}}{\mathrm{ML}^{-3}} = \mathrm{L}^2\mathrm{T}^{-1}$$

Result.

$$\left[\frac{p}{\rho g}\right] = \left[\frac{V^2}{2g}\right] = [\,z\,] = \mathrm{L}, \qquad [\,\mu\,] = \mathrm{ML}^{-1}\mathrm{T}^{-1}, \qquad [\,\nu\,] = \mathrm{L}^2\mathrm{T}^{-1}$$

What it tells us. The dimension check catches errors. If a student writes the pressure head as p/ρ, the dimensions are $\mathrm{L}^2\mathrm{T}^{-2}$, not L, so the factor g is missing. Use this check on any formula you are unsure about.

Common mistakes and exam tips.

- Writing dimensions of force as MLT^{-1}. Force is mass times acceleration, so MLT^{-2}.
- Mixing up the units of μ and ν: μ is in $\mathrm{Pa{\cdot}s}$, ν is in m^2/s.
- Using 1 poise as 1 $\mathrm{Pa{\cdot}s}$. It is $0.1\,\mathrm{Pa\cdot s}$.
- Using 1 bar as $10^4\,\mathrm{N/m}^2$. It is $10^5\,\mathrm{N/m}^2$, and $1\,\mathrm{N/cm}^2 = 10^4\,\mathrm{N/m}^2$.

Check your understanding.

1. CORE What are the dimensions of kinematic viscosity? *($\mathrm{L}^2\mathrm{T}^{-1}$, unit m^2/s.)*
2. CORE Convert 5 cP to $\mathrm{Pa{\cdot}s}$. *($5\times10^{-3}\,\mathrm{Pa\cdot s}$.)*
3. HOTS A student writes the pressure at depth as $p = \rho h$. Is it homogeneous? *(No. $[\,\rho h\,] = \mathrm{ML}^{-2}$, but $[\,p\,] = \mathrm{ML}^{-1}\mathrm{T}^{-2}$. The factor g is missing.)*

3. Density, specific weight, specific volume and specific gravity (L2) CORE

All four describe how much matter a fluid packs into a volume; knowing any one (with g and the density of water) gives the other three.

IN SIMPLE WORDS

These four numbers say how heavy or how packed a fluid is. Density ρ is mass in each cubic metre. Specific weight w is the weight of each cubic metre. Specific volume v is the volume taken by each kilogram, so it is $1/\rho$. Specific gravity S compares a liquid with water: oil with $S = 0.8$ floats on water because it is lighter. If you know one of them, you can find the others.

$$\rho = \frac{m}{V}, \qquad w = \frac{W}{V} = \rho g, \qquad v = \frac{1}{\rho}, \qquad S = \frac{\rho_{liquid}}{\rho_{water}} = \frac{w_{liquid}}{w_{water}}$$

Property	Definition	Water at 4 °C	Remarks
Mass density ρ	Mass per unit volume	1000 kg/m³	Air at 20 °C, 1 atm: 1.2 kg/m³
Specific weight w	Weight per unit volume	9810 N/m³	Changes with g
Specific volume v	Volume per unit mass	$10^{-3}\ \mathrm{m^3/kg}$	Used mostly for gases
Specific gravity S	Ratio to water (gases: to air)	1	Dimensionless; mercury 13.6

For a gas, density follows the equation of state $p = \rho RT$, so it changes strongly with pressure and temperature; for liquids it is nearly constant.

Worked examples

Step-by-step method.

1. Write down what is given (mass, weight, volume or S) and convert it to SI units. For example, 1 litre $= 10^{-3}\ \mathrm{m^3}$.
2. If S is given, find the density from $\rho = S \times 1000\ \mathrm{kg/m^3}$.
3. If mass and volume are given, use $\rho = m/V$. If weight and volume are given, use $w = W/V$ first.
4. Convert between them with $w = \rho g$ and $v = 1/\rho$.
5. Give each answer with its unit. S has no unit.

Example 3.0 (warm-up) CORE A tank holds $0.05\ \mathrm{m^3}$ of oil of mass 40 kg. Find the density, specific weight and specific gravity.

1. $\rho = \dfrac{m}{V} = \dfrac{40}{0.05} = 800\ \mathrm{kg/m^3}$
2. $w = \rho g = 800 \times 9.81 = 7848\ \mathrm{N/m^3}$
3. $S = \dfrac{800}{1000} = 0.8$

Example 3.1 CORE One litre of a liquid weighs 7 N. Find its specific weight, density, specific gravity and specific volume.

1. $w = \dfrac{7}{10^{-3}} = 7000\ \mathrm{N/m^3}$
2. $\rho = \dfrac{w}{g} = \dfrac{7000}{9.81} = 713.6\ \mathrm{kg/m^3}$
3. $S = \dfrac{713.6}{1000} = 0.714$
4. $v = \dfrac{1}{\rho} = 1.401 \times 10^{-3}\ \mathrm{m^3/kg}$

Example 3.2 CORE An oil has specific gravity 0.85. Find its density, specific weight and specific volume.

1. $\rho = 0.85 \times 1000 = 850\ \mathrm{kg/m^3}$
2. $w = 850 \times 9.81 = 8338.5\ \mathrm{N/m^3}$
3. $v = \dfrac{1}{850} = 1.176 \times 10^{-3}\ \mathrm{m^3/kg}$

Common mistakes and exam tips.

- Using mass in kg where weight in N is needed. The weight is $W = mg$.
- Forgetting to change litres to $\mathrm{m^3}$: $1\ \mathrm{L} = 10^{-3}\ \mathrm{m^3}$.
- Giving a unit to S. Specific gravity is a pure ratio.
- Taking S of a gas relative to water. Gases are compared with air.

Check your understanding.

1. CORE A liquid has $\rho = 900\ \mathrm{kg/m^3}$. What is S? *($S = 0.9$.)*
2. CORE Find the specific weight of water. *($w = 1000 \times 9.81 = 9810\ \mathrm{N/m^3}$.)*
3. HOTS Why does the density of a gas change strongly with pressure, but that of a liquid hardly at all? *(A gas obeys $p = \rho RT$ and is compressible; a liquid is nearly incompressible.)*

4. Viscosity (L3) CORE

Viscosity is a fluid's resistance to the relative motion of its layers. It is measured by the dynamic viscosity μ from Newton's law, $\tau = \mu\, du/dy$, and by the kinematic viscosity ν, which is the dynamic viscosity per unit density:

$$\mu = \frac{\tau}{du/dy}, \qquad \nu = \frac{\mu}{\rho}$$

IN SIMPLE WORDS

Viscosity is the “thickness” of a fluid, its resistance to being stirred. Honey has high viscosity and water has low viscosity. Dynamic viscosity μ links the shear stress to the velocity gradient. Kinematic viscosity ν is μ divided by density. It tells how quickly a fluid spreads momentum, and it is the one that appears in the Reynolds number.

Fluid at 20 °C	μ (N·s/m²)	ν (m²/s)
Water	1.0×10^{-3}	1.0×10^{-6}
Air	1.8×10^{-5}	1.5×10^{-5}
SAE 30 oil	about 0.29	about 3.3×10^{-4}
Glycerine	1.49	1.18×10^{-3}

Air has a far lower μ than water but a higher ν, because its density is about 830 times lower.

4.1 Effect of temperature CORE

IN SIMPLE WORDS

Heat a pot of honey and it flows easily. Heat air and it becomes a little more resistant. In a liquid, the molecules cling to each other, and heat loosens this grip. In a gas, the molecules are far apart, and faster molecules swap momentum between layers more quickly. That is why viscosity falls for liquids and rises for gases as temperature goes up.

- **Liquids:** viscosity falls as temperature rises. Viscosity in liquids comes mainly from cohesion between molecules, which weakens when they are heated.
- **Gases:** viscosity rises with temperature. In gases it comes mainly from the exchange of momentum between layers by moving molecules, which speeds up when they are heated.
- Pressure has little effect on viscosity at ordinary pressures.

4.2 Viscous torque and power in a journal bearing CORE

IN SIMPLE WORDS

Inside a bearing, a thin oil film separates the spinning shaft from the fixed housing. The oil layer on the shaft moves with it, and the layer on the housing stays still. The oil is sheared like the two plates in Section 1. This shear gives a drag force on the shaft. Drag times the shaft radius gives a torque, and torque times speed gives the power wasted as heat.

A shaft of diameter D turning at N rpm in a bearing of length L with an oil film of thickness t: the surface speed is $U = \pi DN/60$, the shear stress $\tau = \mu U/t$, the drag force $F = \tau\,(\,\pi DL\,)$, the torque $T = F \times D/2$, and the power lost $P = 2\pi NT/60$.

Derivation from scratch: torque and power lost in a journal bearing

Aim. Prove that $T = \dfrac{\mu U}{t}\ (\pi D L)\ \dfrac{D}{2}$ and $P = \dfrac{2\pi N T}{60}$.

Assumptions.

1. The oil film is thin ($t \ll D$), so it can be treated like the gap between two flat plates.
2. The oil sticks to both surfaces (no slip): its speed is U at the shaft and zero at the fixed bearing.
3. The velocity varies linearly across the film, and the oil is Newtonian with constant μ.
4. The shaft turns steadily at N rpm. Shear at the ends of the shaft is neglected.

Step 1. Surface speed of the shaft. One revolution is 2π rad, and one minute is 60 s. The angular speed is:

$$\omega = \frac{2\pi N}{60}$$

The surface of the shaft is at radius $D/2$, so its speed is:

$$U = \omega \frac{D}{2} = \frac{2\pi N}{60} \times \frac{D}{2} = \frac{\pi D N}{60}$$

Step 2. Velocity gradient in the film. The speed rises from 0 to U across the thickness t, in a straight line.

$$\frac{du}{dy} = \frac{U - 0}{t} = \frac{U}{t}$$

Step 3. Shear stress. Apply Newton's law of viscosity.

$$\tau = \mu \frac{du}{dy} = \frac{\mu U}{t}$$

Step 4. Drag force. The shear acts over the cylindrical surface of the shaft. Its area is the circumference times the length.

$$A_s = \pi D \times L, \qquad F = \tau A_s = \frac{\mu U}{t}\ (\pi D L)$$

The force acts along the surface, against the rotation.

Step 5. Torque. The force acts at the radius $D/2$ from the axis.

$$T = F \times \frac{D}{2} = \frac{\mu U}{t}\ (\pi D L)\ \frac{D}{2}$$

Step 6. Power. Power is torque times angular speed.

$$P = T\omega = T \times \frac{2\pi N}{60}$$

Result.

$$U = \frac{\pi D N}{60}, \qquad T = \frac{\mu U}{t}\ (\pi D L)\ \frac{D}{2}, \qquad P = \frac{2\pi N T}{60}$$

What it tells us. Torque is proportional to μ, to the speed and to the surface area, and inversely proportional to the film thickness. Since $T \propto N$ and $P \propto NT$, the power lost rises with the square of the speed. A thicker oil film or a thinner oil lowers the loss.

Worked examples

Step-by-step method.

1. Convert every length to metres (mm, cm) and speed to m/s; convert viscosity from poise or cP to $\mathrm{Pa{\cdot}s}$.
2. For a plate problem, find the gradient du/dy (or U/t for a thin film), then $\tau = \mu\, du/dy$. If τ is given, solve for μ.
3. For a bearing, find the surface speed $U = \pi DN/60$.
4. Find $\tau = \mu U/t$, then the force $F = \tau \times \pi DL$.
5. Find the torque $T = F \times D/2$ and the power $P = 2\pi NT/60$.
6. Check the units: τ in $\mathrm{N/m^2}$, T in $\mathrm{N{\cdot}m}$, P in W.

Example 4.0 (warm-up) CORE A plate moves at 0.5 m/s over a fixed plate 2 mm away. The oil between them has $\mu = 0.1\ \mathrm{N \cdot s/m^2}$. Find the shear stress.

1. $t = 2\ \mathrm{mm} = 0.002\ \mathrm{m}$, so $\dfrac{du}{dy} = \dfrac{U}{t} = \dfrac{0.5}{0.002} = 250\ \mathrm{s^{-1}}$
2. $\tau = \mu \dfrac{du}{dy} = 0.1 \times 250 = 25\ \mathrm{N/m^2}$

Example 4.1 CORE A plate 0.5 mm from a fixed plate moves at 60 cm/s and needs a force of 120 N per m² of plate area. Find the dynamic viscosity of the oil between them.

1. $\dfrac{du}{dy} = \dfrac{0.6}{0.5 \times 10^{-3}} = 1200\ \mathrm{s^{-1}}$
2. $\mu = \dfrac{\tau}{du/dy} = \dfrac{120}{1200} = 0.1\ \mathrm{N \cdot s/m^2}$ (1 poise)

Example 4.2 CORE The velocity profile over a plate is $u = (2/3)\, y - y^2$, with u in m/s and y in m. For $\mu = 8.63$ poise, find the shear stress at $y = 0$, 0.075 m and 0.15 m.

1. $\mu = 0.863\ \mathrm{N \cdot s/m^2}$; $\dfrac{du}{dy} = \dfrac{2}{3} - 2y$
2. $y = 0$: $\dfrac{du}{dy} = 0.667$, $\tau = 0.575\ \mathrm{N/m^2}$
3. $y = 0.075\ \mathrm{m}$: $\dfrac{du}{dy} = 0.517$, $\tau = 0.446\ \mathrm{N/m^2}$
4. $y = 0.15\ \mathrm{m}$: $\dfrac{du}{dy} = 0.367$, $\tau = 0.316\ \mathrm{N/m^2}$

Example 4.3 CORE A 100 mm shaft turns at 600 rpm in a bearing 150 mm long. The oil film is 0.5 mm thick and $\mu = 0.1\ \mathrm{N \cdot s/m^2}$. Find the torque and the power lost.

1. $U = \dfrac{\pi \times 0.1 \times 600}{60} = 3.142\ \mathrm{m/s}$

2. $\tau = \dfrac{0.1 \times 3.142}{0.0005} = 628.3\ \mathrm{N/m^2}$
3. $F = 628.3 \times \pi \times 0.1 \times 0.15 = 29.61\ \mathrm{N}$; $T = 29.61 \times 0.05 = \ 1.48\ \mathrm{N \cdot m}$
4. $P = \dfrac{2\pi \times 600 \times 1.48}{60} = \ 93.0\ \mathrm{W}$

Common mistakes and exam tips.

- Using the shaft diameter where the radius is needed. The lever arm for torque is $D/2$.
- Leaving the film thickness in mm. $0.5\ \mathrm{mm} = 0.0005\ \mathrm{m}$.
- Using the wrong area. The shear area of a shaft is πDL, not $\pi D^2/4$.
- Forgetting the unit conversion for poise: $8.63\ \mathrm{P} = 0.863\ \mathrm{N \cdot s/m^2}$.
- Saying viscosity of all fluids falls with temperature. It falls for liquids and rises for gases.

Check your understanding.

1. CORE What happens to the viscosity of water when it is heated? *(It falls.)*
2. CORE Express 1 poise in $\mathrm{N \cdot s/m^2}$. *($0.1\ \mathrm{N \cdot s/m^2}$.)*
3. HOTS The shaft speed in a bearing is doubled. What happens to the power lost? *(It becomes four times as large, since $T \propto N$ and $P \propto NT \propto N^2$.)*

5. Compressibility and bulk modulus (L4) CORE

A fluid's volume shrinks when pressure rises. The **bulk modulus** K measures the resistance to this: the pressure rise needed per unit fractional decrease in volume. **Compressibility** is its reciprocal.

IN SIMPLE WORDS

Squeeze a syringe full of water with your thumb on the nozzle. It hardly moves. Squeeze one full of air and the plunger goes in easily. The bulk modulus K measures this stiffness: it is the pressure rise needed to squash the volume by a given fraction. A large K means a stiff fluid. Water is stiff, so we usually treat it as incompressible.

$$K = \ -\frac{dp}{dV/V} = \frac{dp}{d\rho/\rho}, \qquad \beta = \frac{1}{K}$$

The minus sign makes K positive, since volume falls as pressure rises. The second form follows because mass is fixed, so $dV/V = \ -d\rho/\rho$.

Derivation from scratch: bulk modulus relations and the value of K for a gas

Aim. Prove that $K = -\dfrac{dp}{dV/V} = \dfrac{dp}{d\rho/\rho}$, and that $K = p$ for an isothermal gas and $K = kp$ for an adiabatic gas.

Assumptions.

1. A fixed mass m of fluid is compressed, so no fluid enters or leaves.
2. K is defined as the pressure increase per unit fractional decrease in volume.
3. For a gas, $pV = \text{constant}$ (isothermal) or $pV^k = \text{constant}$ (adiabatic).

Step 1. Mass is fixed. Mass equals density times volume.

$$m = \rho V = \text{constant}$$

Step 2. Take logarithms and differentiate.

$$\ln \rho + \ln V = \ln m \quad \Rightarrow \quad \frac{d\rho}{\rho} + \frac{dV}{V} = 0 \quad \Rightarrow \quad \frac{dV}{V} = -\frac{d\rho}{\rho}$$

Step 3. Put this in the definition. The minus sign cancels.

$$K = -\frac{dp}{dV/V} = -\frac{dp}{-d\rho/\rho} = \frac{dp}{d\rho/\rho}$$

Step 4. Isothermal gas. Differentiate $pV = C$ with the product rule.

$$p\,dV + V dp = 0 \quad \Rightarrow \quad dp = -p\frac{dV}{V} \quad \Rightarrow \quad K = -\frac{dp}{dV/V} = p$$

Step 5. Adiabatic gas. Differentiate $pV^k = C$.

$$V^k\,dp + k\,p\,V^{k-1}\,dV = 0$$

Divide by V^k:

$$dp + k\,p\frac{dV}{V} = 0 \quad \Rightarrow \quad dp = -k\,p\frac{dV}{V} \quad \Rightarrow \quad K = kp$$

Result.

$$K = -\frac{dp}{dV/V} = \frac{dp}{d\rho/\rho}, \qquad K_{\text{isothermal}} = p, \qquad K_{\text{adiabatic}} = kp$$

What it tells us. A volume fall of 1% is the same as a density rise of 1%. A gas becomes stiffer as its pressure rises. An adiabatic gas is k times stiffer than an isothermal one, because it also heats up as it is squeezed.

Fluid	Bulk modulus K	Remarks
Water	about 2.1–$2.2 \times 10^9\ \mathrm{N/m^2}$	1% volume change needs about 220 bar
Mercury	about $2.6 \times 10^{10}\ \mathrm{N/m^2}$	Nearly incompressible
Gas, isothermal ($pV = \text{const}$)	$K = p$	Depends on pressure
Gas, adiabatic ($pV^k = \text{const}$)	$K = kp$	$k = 1.4$ for air

Liquids are treated as incompressible in most problems; compressibility matters for water hammer (Unit III), hydraulic presses and high-speed gas flow. The speed of sound in a fluid is $\sqrt{K/\rho}$: about 1450 m/s in water and 343 m/s in air at 20 °C.

Worked examples

Step-by-step method.

1. Write the pressure rise dp in $\mathrm{N/m^2}$ ($1\ \mathrm{N/cm^2} = 10^4\ \mathrm{N/m^2}$, $1\ \mathrm{bar} = 10^5\ \mathrm{N/m^2}$).
2. Write the fractional volume change dV/V as a decimal (0.15% is 0.0015). A fall in volume has a negative dV.
3. Use $K = dp/(-dV/V)$, or $\dfrac{dV}{V} = \dfrac{dp}{K}$ if K is given.
4. Find the change in volume: $\Delta V = (dV/V) \times V$.
5. For a gas use $K = p$ (isothermal) or $K = kp$ (adiabatic), with p absolute.

Example 5.0 (warm-up) CORE An oil has $K = 1.5 \times 10^9\ \mathrm{N/m^2}$. Its pressure rises by $3\ \mathrm{MN/m^2}$. Find the fractional decrease in volume.

1. $dp = 3 \times 10^6\ \mathrm{N/m^2}$
2. $\dfrac{dV}{V} = \dfrac{dp}{K} = \dfrac{3 \times 10^6}{1.5 \times 10^9} = 0.002\ (0.2\%)$

Example 5.1 CORE The pressure of a liquid rises from 70 N/cm² to 370 N/cm² and its volume falls by 0.15%. Find the bulk modulus.

1. $dp = (370 - 70) \times 10^4 = 3 \times 10^6\ \mathrm{N/m^2}$; $dV/V = -0.0015$
2. $K = \dfrac{3 \times 10^6}{0.0015} = 2 \times 10^9\ \mathrm{N/m^2}$

Example 5.2 CORE Find the decrease in volume of 1 m³ of water ($K = 2.1 \times 10^9\ \mathrm{N/m^2}$) when the pressure rises by 10 MN/m².

1. $\dfrac{dV}{V} = \dfrac{dp}{K} = \dfrac{10 \times 10^6}{2.1 \times 10^9} = 0.00476$
2. $\text{Decrease} = 0.00476 \times 1 = 0.00476\ \mathrm{m^3}\ (4.76\ \text{litres})$

Common mistakes and exam tips.

- Dropping the minus sign and then getting a negative K. The minus sign is there because volume falls when pressure rises.
- Mixing percent and decimal: 1% is 0.01.
- Leaving pressure in $\mathrm{N/cm^2}$ or bar. Convert to $\mathrm{N/m^2}$ first.
- Using gauge pressure for a gas K. Use the absolute pressure.

Check your understanding.

1. CORE What does a large bulk modulus mean? *(The fluid is hard to compress.)*
2. CORE $K = 2 \times 10^9\ \mathrm{N/m^2}$ and $dp = 2 \times 10^6\ \mathrm{N/m^2}$. Find $|dV/V|$. *(0.001, that is 0.1%.)*
3. HOTS Which is more compressible, water ($K \approx 2.1 \times 10^9$) or mercury ($K \approx 2.6 \times 10^{10}$)? *(Water, because it has the smaller K.)*

6. Surface tension and capillarity (L4–L5) CORE

Molecules at a liquid's free surface are pulled inward by cohesion with no balancing pull from above, so the surface behaves like a stretched membrane. **Surface tension** σ is the tensile force per unit length of a line on that surface (N/m). Water–air at 20 °C: $\sigma = 0.0725\ \mathrm{N/m}$; mercury–air: about 0.52 N/m. It falls as temperature rises.

IN SIMPLE WORDS

A water strider walks on a pond because the surface acts like a thin stretched skin. Molecules at the surface are pulled inward by the ones below, so the surface tries to shrink. Surface tension σ is the pull of this skin, in newtons for each metre of length. It makes small drops round, makes bubbles hold extra pressure, and lifts water up narrow tubes.

6.1 Pressure inside a droplet, bubble and jet CORE

IN SIMPLE WORDS

The skin of a drop or bubble squeezes the liquid inside, like the rubber of a balloon. So the pressure inside is higher than outside. A smaller drop is squeezed harder, because the skin is more tightly curved. A soap bubble has an inner and an outer skin, so it holds twice the excess of a drop of the same size.

Cutting the drop in half and balancing the pressure force on the cut face against the surface tension round its rim gives the excess of inside over outside pressure:

Shape	Force balance	Pressure excess p
Liquid droplet (one surface)	$p\left(\pi d^2/4\right) = \sigma\,(\pi d)$	$4\sigma/d$
Soap bubble (two surfaces)	$p\left(\pi d^2/4\right) = 2\sigma\,(\pi d)$	$8\sigma/d$
Liquid jet (cylinder, length L)	$p\,(Ld) = \sigma\,(2L)$	$2\sigma/d$

Derivation from scratch: pressure inside a droplet, a soap bubble and a liquid jet

Aim. Prove that the excess pressure inside is $4\sigma/d$ for a droplet, $8\sigma/d$ for a soap bubble and $2\sigma/d$ for a jet.

Assumptions.

1. The droplet and bubble are spheres of diameter d. The jet is a long circular cylinder of diameter d.
2. p is the excess of the inside pressure over the outside pressure, and it is the same everywhere inside.
3. The surface tension σ is the same everywhere. The weight of the liquid and the thickness of the soap film are neglected.
4. A soap film has two surfaces (inside and outside), and each one pulls with σ.

Step 1. Droplet: the free body. Cut the sphere through its centre by a plane. Look at one half (a hemisphere). The excess pressure pushes this half away from the other half. The net push of a pressure on a curved surface equals the pressure times the projected area, which is the circular cut face of area $\pi d^2/4$.

$$F_p = p \times \frac{\pi d^2}{4}$$

Step 2. Droplet: surface tension. The surface tension acts along the rim of the cut, whose length is the circumference πd. It pulls the hemisphere back toward the other half.

$$F_\sigma = \sigma \times \pi d$$

Step 3. Droplet: balance. The hemisphere is in equilibrium.

$$p\,\frac{\pi d^2}{4} = \sigma\,\pi d \quad \Rightarrow \quad p = \sigma\,\pi d \times \frac{4}{\pi d^2} = \frac{4\sigma}{d}$$

Step 4. Soap bubble. The pressure force is the same, but the film has two surfaces, so the rim force doubles.

$$p\,\frac{\pi d^2}{4} = 2\sigma\,\pi d \quad \Rightarrow \quad p = \frac{8\sigma}{d}$$

Step 5. Jet. Take a length L of the jet and cut it lengthwise through the axis. For one half, the projected area of the cut face is the rectangle $L \times d$. The surface tension pulls along the two straight edges, each of length L.

$$p\ (\ Ld\ )\ = \sigma\ (\ 2L\ ) \quad \Rightarrow \quad p = \frac{2\sigma}{d}$$

Result.

$$p_{\text{droplet}} = \frac{4\sigma}{d}, \qquad p_{\text{bubble}} = \frac{8\sigma}{d}, \qquad p_{\text{jet}} = \frac{2\sigma}{d}$$

What it tells us. The excess pressure is inversely proportional to the diameter, so tiny drops hold very high pressure. For the same d, the bubble has twice the pressure of the droplet, and the jet has half.

6.2 Capillarity CORE

IN SIMPLE WORDS

Dip a thin glass tube in water and the water climbs inside it, a little above the outside level. The skin of the water grips the glass and pulls the water up. The narrower the tube, the higher the rise. Mercury does the opposite: it pulls away from the glass, and its level inside the tube is lower. The contact angle θ tells which one happens.

In a narrow tube dipped in a liquid, the level inside rises or falls compared with the outside. The vertical component of surface tension round the tube's perimeter balances the weight of the raised column:

$$\sigma \cos\theta \ (\ \pi d \ ) \ = \rho g \frac{\pi d^2}{4} h \quad \Rightarrow \quad h = \frac{4\sigma \cos\theta}{\rho g d}$$

Derivation from scratch: capillary rise or depression

Aim. Prove that the rise (or fall) of liquid in a tube of diameter d is $h = \dfrac{4\sigma \cos\theta}{\rho g d}$.

Assumptions.

1. The tube is clean and vertical, with inside diameter d. The liquid meets the wall at a contact angle θ, measured through the liquid.
2. The liquid in the tube is at rest, and the small volume of the curved meniscus is neglected, so the raised liquid is a cylinder of height h.
3. The atmosphere acts on the outer free surface and on the meniscus equally, so it cancels.

Step 1. Upward pull of the surface tension. The meniscus meets the wall along a circle of length πd. Surface tension pulls along the liquid surface with σ per unit length, at angle θ to the wall. Its vertical component is $\sigma \cos\theta$ per unit length.

$$F_{\text{up}} = \sigma \cos\theta \times \pi d$$

Step 2. Weight of the raised column. The volume is the tube area times the height.

$$W = \rho g \times \frac{\pi d^2}{4} h$$

Step 3. Equilibrium. The column is at rest, so the upward pull equals the weight.

$$\sigma \cos\theta \, \pi d = \rho g \frac{\pi d^2}{4} h$$

Step 4. Solve for h. Cancel π and one factor of d.

$$h = \frac{\sigma \cos\theta \, \pi d \times 4}{\rho g \, \pi d^2} = \frac{4\sigma \cos\theta}{\rho g d}$$

Result.

$$h = \frac{4\sigma \cos\theta}{\rho g d}$$

What it tells us. The rise is inversely proportional to the tube diameter: halving d doubles h. If $\theta < 90^\circ$ (water on glass), $\cos\theta > 0$ and the liquid rises. If $\theta > 90^\circ$ (mercury on glass), $\cos\theta < 0$, the vertical pull acts downward, and h is negative: the liquid is depressed.

- **Capillary rise:** liquids that wet the glass (adhesion > cohesion), $\theta < 90^\circ$. For water and clean glass, $\theta \approx 0$.
- **Capillary depression:** liquids that do not wet the glass (cohesion > adhesion), $\theta > 90^\circ$, so h is negative. For mercury and glass, $\theta \approx 130^\circ$.
- Capillarity explains why manometer and piezometer tubes should be wider than about 6 mm, and drives the rise of water in soil and wicks.

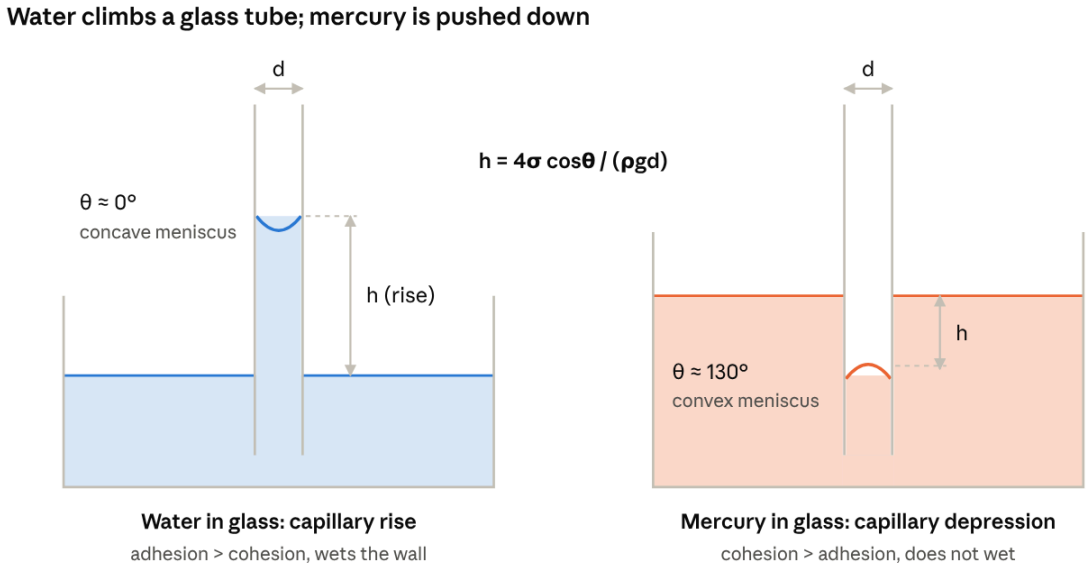


Figure 2. Capillary rise of water and capillary depression of mercury

The same formula covers both cases: $\cos\theta$ is positive for water (rise) and negative for mercury (depression).

Worked examples

Step-by-step method.

1. Convert the diameter to metres and look up σ (and θ for capillarity).
2. Decide the shape: a droplet has one surface, a soap bubble has two, a jet is a cylinder.
3. Use $p = 4\sigma/d$ (droplet), $p = 8\sigma/d$ (bubble) or $p = 2\sigma/d$ (jet).
4. For capillarity use $h = 4\sigma\cos\theta/(\rho g d)$ with the density ρ of the liquid in the tube (mercury: $13\,600$).
5. Report the sign: positive h is a rise, negative h is a depression.

Example 6.0 (warm-up) CORE Find the excess pressure inside a soap bubble of diameter 20 mm. Take $\sigma = 0.05\ \mathrm{N/m}$.

1. $d = 20\ \mathrm{mm} = 0.02\ \mathrm{m}$; a bubble has two surfaces, so $p = \dfrac{8\sigma}{d}$
2. $p = \dfrac{8 \times 0.05}{0.02} = 20\ \mathrm{N/m^2}$

Example 6.1 CORE Find the pressure excess inside a water droplet of diameter 0.04 mm ($\sigma = 0.0725\ \mathrm{N/m}$).

$$p = \frac{4\sigma}{d} = \frac{4 \times 0.0725}{0.04 \times 10^{-3}} = 7250\ \mathrm{N/m^2}$$

Example 6.2 CORE Find the capillary rise of water and the capillary depression of mercury in a glass tube of 2.5 mm diameter. Water: $\sigma = 0.0725\ \mathrm{N/m}$, $\theta = 0$. Mercury: $\sigma = 0.52\ \mathrm{N/m}$, $\theta = 130^\circ$, $S = 13.6$.

1. Water: $h = \dfrac{4 \times 0.0725}{1000 \times 9.81 \times 0.0025} = 11.8\ \mathrm{mm}\ \text{(rise)}$
2. Mercury: $h = \dfrac{4 \times 0.52 \times \cos 130^\circ}{13\,600 \times 9.81 \times 0.0025} = -4.0\ \mathrm{mm}\ \text{(depression)}$

Common mistakes and exam tips.

- Using $4\sigma/d$ for a soap bubble. A bubble has two surfaces, so use $8\sigma/d$.
- Putting the diameter in mm. A 0.04 mm droplet is 0.04×10^{-3} m.
- Using the density of water for mercury in capillarity. Use $13\,600\ \mathrm{kg/m^3}$.
- Dropping the sign for mercury. $\cos 130^\circ$ is negative, so h is a depression.
- Confusing excess pressure with absolute pressure inside the drop. The formulas give the excess over the outside.

Check your understanding.

1. CORE Which is larger, the excess pressure in a droplet or in a bubble of the same diameter? *(The bubble: $8\sigma/d$ against $4\sigma/d$.)*

2. CORE The tube diameter is halved. What happens to the capillary rise? *(It doubles, since $h \propto 1/d$.)*
3. HOTS Why does the mercury level fall in a glass tube? *(Mercury does not wet glass, $\theta > 90^\circ$, so $\cos\theta < 0$.)*

7. Incompressible flow and Bernoulli's equation (L6) CORE

In steady, frictionless, incompressible flow, the sum of pressure head, velocity head and datum head is the same at every point on a streamline. This is Bernoulli's equation, the energy equation behind all three flow meters in this unit.

> **IN SIMPLE WORDS**
> Bernoulli's equation is the energy bookkeeping of a flowing liquid. Water can carry energy in three ways: its pressure, its speed and its height. Along a streamline, if there is no friction, the total stays constant, so one kind can turn into another. When a garden hose is squeezed at the nozzle, the water speeds up and its pressure drops. The three flow meters in this unit use exactly this effect.

7.1 Incompressible flow CORE

> **IN SIMPLE WORDS**
> A flow is called incompressible when the density does not change. Water in pipes behaves this way. Air behaves this way too if it moves much slower than sound. Then the same volume of fluid that enters a pipe also leaves it, and we can take ρ as a constant in the formulas.

A flow is treated as **incompressible** when density changes are negligible: $\rho = \text{constant}$. This holds for liquids at ordinary pressures, and for gases when the Mach number (V divided by the speed of sound) is below about 0.3, where density changes by less than about 5%.

7.2 Euler's equation of motion CORE

> **IN SIMPLE WORDS**
> Euler's equation is just Newton's second law, force equals mass times acceleration, written for a tiny slug of fluid sliding along a streamline. The forces are the pressure on its two ends and its weight. Friction is ignored. The result tells how pressure, height and speed change together from one point to the next.

Derivation from scratch: Euler's equation of motion along a streamline

Aim. Prove that for flow along a streamline $\dfrac{dp}{\rho} + g\,dz + v\,dv = 0$.

Assumptions.

1. The fluid is ideal: viscous forces are zero. Only pressure and gravity act.
2. The flow is steady, so the velocity at a point does not change with time.
3. The element is small, so the pressure is uniform over its cross-section.
4. The element moves along a streamline; s is the distance along it.

Step 1. The element. Take a small cylinder of cross-section dA and length ds along the streamline. Its volume is $dA\,ds$, its mass is $\rho\,dA\,ds$, and its weight is $\rho g\,dA\,ds$. Let the streamline make an angle θ with the vertical, so a step ds along it raises the element by $dz = ds\cos\theta$. (In this box θ is this angle from the vertical, not the contact angle of Section 6, and v is the speed along the streamline, the same quantity as V elsewhere, not the specific volume.)

Step 2. Pressure forces along s. The pressure at the upstream face is p, and at the downstream face it is $p + \dfrac{\partial p}{\partial s}ds$. The upstream face pushes the element forward, the downstream face pushes it backward.

$$F_{\text{up face}} = p\,dA, \qquad F_{\text{down face}} = \left(p + \frac{\partial p}{\partial s}ds\right)dA$$

The pressure on the curved sides acts at right angles to the streamline, so it adds nothing along s.

Step 3. Weight along s. The component of the weight along the streamline is the weight times $\cos\theta$. It points backward when the flow goes upward.

$$W_s = \rho g\,dA\,ds\,\cos\theta$$

Step 4. Newton's second law along s. Net force in the flow direction equals mass times acceleration a_s.

$$p\,dA - \left(p + \frac{\partial p}{\partial s}ds\right)dA - \rho g\,dA\,ds\,\cos\theta = \rho\,dA\,ds\,a_s$$

Step 5. Acceleration. In steady flow the speed v depends only on position s. The element's acceleration is the rate of change of speed with time.

$$a_s = \frac{dv}{dt} = \frac{dv}{ds}\frac{ds}{dt} = v\frac{dv}{ds}$$

Step 6. Simplify the forces. The terms $p\,dA$ cancel. Replace $\cos\theta$ by dz/ds.

$$-\frac{\partial p}{\partial s}\,ds\,dA - \rho g\,dA\,ds\frac{dz}{ds} = \rho\,dA\,ds\,v\frac{dv}{ds}$$

Step 7. Divide by $\rho\,dA\,ds$.

$$-\frac{1}{\rho}\frac{\partial p}{\partial s} - g\frac{dz}{ds} = v\frac{dv}{ds}$$

Step 8. Multiply by ds and collect terms. Note that $\frac{\partial p}{\partial s} ds = dp$ along the streamline.

$$-\frac{dp}{\rho} - g\,dz = v\,dv$$

Result.

$$\frac{dp}{\rho} + g\,dz + v\,dv = 0$$

What it tells us. Along a streamline, a rise in pressure must be paid for by a fall in speed or height. This is Newton's second law in a form that needs no friction term. It is called **Euler's equation**.

7.3 Bernoulli's equation CORE

IN SIMPLE WORDS

Bernoulli's equation says that the three heads add up to the same total at every point on a streamline. The pressure head is the height the pressure could lift the liquid. The velocity head is the height the moving liquid could climb if its speed were turned into height. The datum head is the actual height. When one goes up, another goes down. For a real fluid, friction removes some total head, which is the loss h_L.

Integrating Euler's equation with ρ constant and dividing by g:

$$\frac{p}{\rho g} + \frac{V^2}{2g} + z = \text{constant} \qquad \text{or} \qquad \frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

Each term is energy per unit weight of fluid, in metres: **pressure head** $p/(\rho g)$, **velocity (kinetic) head** $V^2/(2g)$ and **datum (potential) head** z. Their sum is the total head.

Assumptions: (1) ideal fluid, no viscosity; (2) steady flow; (3) incompressible; (4) along a streamline (or irrotational flow); (5) no shaft work or heat transfer.

Derivation from scratch: Bernoulli's equation from Euler's equation

Aim. Prove that $\frac{p}{\rho g} + \frac{V^2}{2g} + z$ is constant along a streamline.

Assumptions.

1. The fluid is ideal (no viscosity).
2. The flow is steady.
3. The fluid is incompressible, so ρ is constant and can be taken outside an integral.
4. The points lie on one streamline (or the flow is irrotational).
5. There is no shaft work and no heat transfer.

Step 1. Start from Euler's equation. Assumptions 1, 2 and 4 are those used to derive it.

$$\frac{dp}{\rho} + g\,dz + v\,dv = 0$$

Step 2. Integrate each term along the streamline. Because ρ and g are constant, each term integrates directly. C is the constant of integration.

$$\frac{1}{\rho}\int dp + g\int dz + \int v\,dv = C \quad \Rightarrow \quad \frac{p}{\rho} + gz + \frac{v^2}{2} = C$$

Step 3. Divide by g. Write V for the speed.

$$\frac{p}{\rho g} + z + \frac{V^2}{2g} = \frac{C}{g} = \text{constant}$$

Step 4. Two points on the same streamline. The constant is the same at both points.

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

Result.

$$\frac{p}{\rho g} + \frac{V^2}{2g} + z = \text{constant}$$

What it tells us. The total head is conserved in ideal flow. If a pipe narrows and V rises, $p/\rho g$ must fall (for the same level). Assumption 3 is needed so that $\int dp/\rho$ becomes p/ρ.

Real fluids. Friction converts part of the energy to heat, so the total head falls along the flow by the head loss h_L:

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + h_L$$

With the continuity equation $Q = A_1V_1 = A_2V_2$, Bernoulli's equation shows that where a pipe narrows the velocity rises and the pressure falls. The venturimeter, orifice meter and Pitot tube all measure flow by this pressure change.

Worked examples

Step-by-step method.

1. Sketch the flow. Choose section 1 and section 2 on a streamline, and a datum level.
2. Find the velocities from continuity: $V = Q/A$ with $A = \pi D^2/4$ (diameter in metres).
3. Write Bernoulli's equation with the three heads at each section, plus h_L if there is a loss.
4. Use the same pressure scale (gauge or absolute) at both sections, and the heads of the same liquid.
5. Solve for the unknown head, then convert to pressure with $p = \rho g h$.
6. Check that the answer is reasonable: the pressure should fall where the velocity rises.

Example 7.0 (warm-up) CORE Water flows through a horizontal pipe. At section 1, $V_1 = 2\ \text{m/s}$ and $p_1 = 100\ \text{kN/m}^2$. At section 2 the pipe is narrower and $V_2 = 6\ \text{m/s}$. Neglecting losses, find p_2.

1. Horizontal pipe, so $z_1 = z_2$. Multiplying Bernoulli's equation by ρg gives $p_1 + \frac{1}{2}\rho V_1^2 = p_2 + \frac{1}{2}\rho V_2^2$.
2. $p_2 = p_1 + \frac{1}{2}\rho\left(V_1^2 - V_2^2\right) = 100\,000 + \frac{1}{2} \times 1000 \times \left(2^2 - 6^2\right)$
3. $p_2 = 100\,000 - 16\,000 = \ 84\,000\ \text{N/m}^2\ \ (\,84\ \text{kPa}\,)$

Example 7.1 CORE Water flows in a pipe of 200 mm diameter at 3 m/s, with a pressure of 29.43 N/cm², at a point 5 m above datum. Find the total head.

1. $\text{Pressure head} = \dfrac{29.43 \times 10^4}{1000 \times 9.81} = 30\ \text{m}$
2. $\text{Velocity head} = \dfrac{3^2}{2 \times 9.81} = 0.459\ \text{m}$; $\text{datum head} = 5\ \text{m}$
3. $\text{Total head} = 30 + 0.459 + 5 = \ 35.46\ \text{m}$

Example 7.2 HOTS A pipe carries 0.05 m³/s of water from section 1 (200 mm diameter, 400 kN/m²) to section 2 (100 mm diameter), 3 m higher. Neglecting losses, find the pressure at section 2.

1. $V_1 = \dfrac{0.05}{(\,\pi/4\,) \times 0.2^2} = 1.592\ \text{m/s}$; $V_2 = \dfrac{0.05}{(\,\pi/4\,) \times 0.1^2} = 6.366\ \text{m/s}$
2. $\dfrac{p_2}{\rho g} = \dfrac{400\,000}{9810} + \dfrac{1.592^2 - 6.366^2}{19.62} + 0 - 3 = 40.77 - 1.94 - 3 = 35.84\ \text{m}$
3. $p_2 = 35.84 \times 9810 = \ 351.6\ \text{kN/m}^2$

Common mistakes and exam tips.

- Using the wrong velocity. Find V from Q/A at each section, with the diameter in metres.
- Mixing gauge and absolute pressure between sections. A jet open to the air has $p = 0$ gauge.
- Forgetting the datum head z when the pipe is inclined or a tank surface is above the outlet.
- Using Bernoulli's equation across a pump, a turbine or a large loss without adding the work or the loss term.
- Dividing pressure by 1000×9.81 for oil. Use $\rho_{\text{oil}}\, g$.

Check your understanding.

1. CORE Name the three heads in Bernoulli's equation. *(Pressure head $p/\rho g$, velocity head $V^2/2g$ and datum head z.)*
2. CORE A horizontal pipe narrows. What happens to the pressure? *(It falls, because the velocity rises.)*
3. HOTS How does the total head at section 2 compare with that at section 1 for a real fluid? *(It is lower by the head loss h_L.)*

8. Venturimeter (L7) CORE

A venturimeter measures the discharge in a pipe from the pressure drop between the pipe and a smooth throat. Its gentle cones keep losses small, so its coefficient of discharge is high, 0.95 to 0.99.

IN SIMPLE WORDS

A venturimeter is a pipe section that narrows and then widens smoothly, like a bottle neck. Where the pipe narrows, the water speeds up and its pressure drops. Measure the drop in pressure and you can work out how much water flows. It is like reading the speed of a river from how much it dips at a narrow gap. The coefficient C_d corrects the ideal answer for small losses.

8.1 Construction CORE

IN SIMPLE WORDS

The meter has three parts: a short cone that narrows the flow, a narrow throat where the pressure is lowest, and a long gentle cone that widens again. The long exit cone lets the water slow down smoothly. Then it does not break away from the wall, and most of the pressure comes back.

1. **Converging cone:** short, included angle about 20–22°; flow accelerates.
2. **Throat:** shortest, smallest section, diameter usually $1/3$ to $3/4$ of the pipe diameter; pressure is lowest here.

3. **Diverging cone:** long, included angle about 5–7°, so the flow slows gradually without separating and most of the pressure is recovered.

A differential manometer is connected between the inlet (section 1) and the throat (section 2).

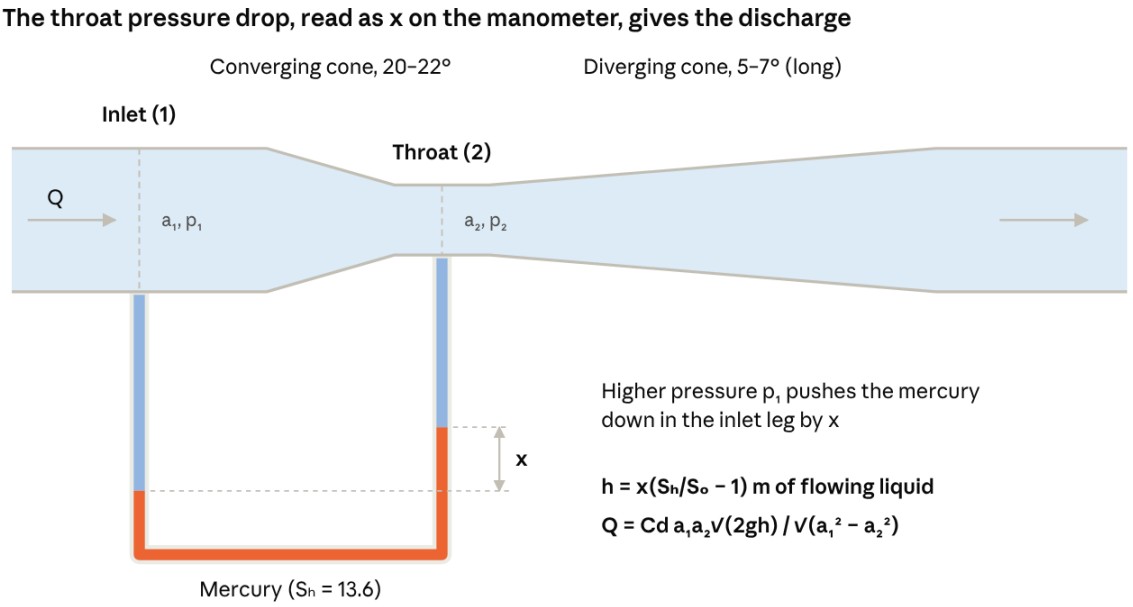


Figure 3. Horizontal venturimeter with a mercury differential manometer

8.2 Discharge (derivation) CORE

IN SIMPLE WORDS

To find the discharge we use two facts together. Bernoulli's equation links the pressure drop to the rise in speed. Continuity links the two speeds to the two areas. Eliminate the inlet speed and the throat speed comes out in terms of the pressure drop. Multiply it by the throat area to get the discharge Q.

Derivation from scratch: discharge through a venturimeter

Aim. Prove that $Q = C_d \dfrac{a_1 a_2 \sqrt{2gh}}{\sqrt{a_1^2 - a_2^2}}$.

Assumptions.

1. The flow is steady and incompressible, and the liquid is treated as ideal between sections 1 and 2. The losses are included later through C_d.
2. Each section has uniform velocity and pressure. Section 1 is at the inlet (area a_1), section 2 at the throat (area a_2), with $a_1 > a_2$.
3. The meter may be horizontal or inclined; z_1 and z_2 are the heights of the section centres. Let h be the difference of piezometric heads (pressure head plus datum head).

Step 1. Bernoulli's equation between 1 and 2.

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

Step 2. Move the pressure and datum terms to one side. Define h and note that for a horizontal meter $z_1 = z_2$, so $h = \left(p_1 - p_2\right) / \left(\rho g\right)$.

$$h = \left(\frac{p_1}{\rho g} + z_1\right) - \left(\frac{p_2}{\rho g} + z_2\right) = \frac{V_2^2 - V_1^2}{2g}$$

Step 3. Continuity. The same discharge passes both sections.

$$a_1 V_1 = a_2 V_2 \quad \Rightarrow \quad V_1 = \frac{a_2}{a_1} V_2$$

Step 4. Eliminate V_1. Substitute into Step 2 and multiply by $2g$.

$$V_2^2 - \frac{a_2^2}{a_1^2} V_2^2 = 2gh \quad \Rightarrow \quad V_2^2 \left(1 - \frac{a_2^2}{a_1^2}\right) = 2gh$$

Step 5. Solve for V_2. Write the bracket with a common denominator, $\frac{a_1^2 - a_2^2}{a_1^2}$.

$$V_2^2 = \frac{2gh\, a_1^2}{a_1^2 - a_2^2} \quad \Rightarrow \quad V_2 = \frac{a_1 \sqrt{2gh}}{\sqrt{a_1^2 - a_2^2}}$$

Step 6. Theoretical discharge. Multiply by the throat area.

$$Q_{th} = a_2 V_2 = \frac{a_1 a_2 \sqrt{2gh}}{\sqrt{a_1^2 - a_2^2}}$$

Step 7. Coefficient of discharge. Friction between the inlet and the throat makes the real discharge smaller. The ratio of actual to theoretical discharge is C_d (0.95 to 0.99).

Result.

$$Q_{th} = \frac{a_1 a_2 \sqrt{2gh}}{\sqrt{a_1^2 - a_2^2}}, \qquad Q = C_d \frac{a_1 a_2 \sqrt{2gh}}{\sqrt{a_1^2 - a_2^2}}$$

What it tells us. For a given meter, $Q \propto \sqrt{h}$: to double the flow, the head difference must rise four times. The same formula holds for an inclined or vertical meter, as long as h is the difference of piezometric heads.

C_d is the coefficient of discharge, which allows for friction between the inlet and throat.

8.3 Finding h from the manometer CORE

IN SIMPLE WORDS

The pressure difference is not read directly. A U-tube filled with a heavy liquid such as mercury is connected between the two tappings. The higher pressure at the inlet pushes the mercury down on one side and up on the other. The height difference x of the mercury tells the pressure difference. We then express this difference as a head of the liquid in the pipe, to put it in the formula.

h is the difference in **pressure heads expressed in the flowing liquid** (specific gravity S_o), from a manometer reading x:

Manometer liquid	h
Heavier than the flowing liquid (S_h), e.g. mercury	$h = x\left(S_h/S_o - 1\right)$
Lighter than the flowing liquid (S_l), inverted U-tube	$h = x\left(1 - S_l/S_o\right)$
Inclined or vertical venturimeter	$h = \left(p_1/\rho g + z_1\right) - \left(p_2/\rho g + z_2\right)$; the manometer formulas above still give it directly

For water with a mercury manometer, $h = 12.6x$.

Derivation from scratch: head difference h from a differential manometer

Aim. Prove that $h = x\left(\dfrac{S_h}{S_o} - 1\right)$ for a heavier manometer liquid and $h = x\left(1 - \dfrac{S_l}{S_o}\right)$ for a lighter one, whether the meter is horizontal or inclined.

Assumptions.

1. The pipe liquid has density ρ_o (specific gravity S_o) and fills the connecting tubes. The manometer liquid does not mix with it.
2. All liquids are at rest in the manometer, so pressure changes only by $\rho g \times$ height.
3. h is the difference of piezometric heads, $h = \left(\dfrac{p_1}{\rho_o g} + z_1\right) - \left(\dfrac{p_2}{\rho_o g} + z_2\right)$, where z_1 and z_2 are the heights of the section centres above a chosen level.

Case 1: heavier manometer liquid (U-tube below, density ρ_h).

Step 1. Choose the level. Let A–A be the mercury surface in the limb connected to section 1. Section 1 is at height z_1 above A–A, and section 2 at height z_2 above A–A. The mercury in the other limb stands higher by the reading x.

Step 2. Pressure at A–A through the limb on the section 1 side. Only pipe liquid of height z_1 lies above A–A.

$$p_{\mathrm{A}} = p_1 + \rho_o g z_1$$

Step 3. Pressure at the same level through the other limb. There is mercury of height x above this level, and then pipe liquid of height $z_2 - x$ up to section 2.

$$p_{\mathrm{A}} = p_2 + \rho_o g\,(\,z_2 - x\,) + \rho_h g\,x$$

Step 4. Equate and rearrange.

$$p_1 + \rho_o g z_1 = p_2 + \rho_o g z_2 - \rho_o g x + \rho_h g x$$

$$\left(p_1 + \rho_o g z_1\right) - \left(p_2 + \rho_o g z_2\right) = \left(\rho_h - \rho_o\right) g\,x$$

Step 5. Divide by $\rho_o g$. The left side is h, and $\rho_h/\rho_o = S_h/S_o$.

$$h = x\left(\frac{\rho_h}{\rho_o} - 1\right) = x\left(\frac{S_h}{S_o} - 1\right)$$

Case 2: lighter manometer liquid (inverted U-tube above, density ρ_l).

Step 6. Choose the level. The higher pressure is at section 1, so the light liquid stands higher by x in the limb on that side. Let B–B be the lower interface (section 2 side). Section 2 is at height z_2 above B–B, and section 1 at height z_1 above B–B. Because the inverted U-tube sits above the pipe, both sections lie below B–B, so z_1 and z_2 are negative here.

Step 7. Pressure at the lower interface. Going up from the section 2 centre to B–B is a rise of $-z_2$ through pipe liquid, which lowers the pressure by $\rho_o g\,(\,-z_2\,)$.

$$p_{\mathrm{B}} = p_2 + \rho_o g z_2$$

Step 8. Pressure at the upper interface. Moving up through the light liquid by x lowers the pressure by $\rho_l g x$. The same pressure is found by going from section 1 up to the upper interface, a rise of $x - z_1$ through the pipe liquid.

$$p_{\mathrm{B}} - \rho_l g x = p_1 - \rho_o g\,(\,x - z_1\,)$$

Step 9. Rearrange. Substitute p_{B} from Step 7 and collect terms.

$$p_2 + \rho_o g z_2 - \rho_l g x = p_1 + \rho_o g z_1 - \rho_o g x$$

$$\left(p_1 + \rho_o g z_1\right) - \left(p_2 + \rho_o g z_2\right) = \left(\rho_o - \rho_l\right) g\,x$$

Step 10. Divide by $\rho_o g$.

$$h = x\left(1 - \frac{\rho_l}{\rho_o}\right) = x\left(1 - \frac{S_l}{S_o}\right)$$

Result.

$$h = x\left(\frac{S_h}{S_o} - 1\right)\ (\text{heavier liquid}), \qquad h = x\left(1 - \frac{S_l}{S_o}\right)\ (\text{lighter liquid})$$

What it tells us. The heights z_1 and z_2 cancel, so the same formula works for horizontal, inclined and vertical meters: the manometer reads the piezometric head difference directly. For water with a mercury manometer, $h = x\,(\,13.6 - 1\,) = 12.6x$.

Worked example

Step-by-step method.

1. Convert the diameters to metres and find $a_1 = \pi D_1^2/4$ and $a_2 = \pi D_2^2/4$.
2. Find h from the manometer with the correct formula, using S_o of the liquid flowing in the pipe.
3. Find $\sqrt{2gh}$ and $\sqrt{a_1^2 - a_2^2}$ separately.
4. Compute $Q_{th} = a_1 a_2 \sqrt{2gh} / \sqrt{a_1^2 - a_2^2}$.
5. Multiply by C_d to get the actual Q. Give the answer in $\mathrm{m^3/s}$ and, if useful, in L/s.

Example 8.0 (warm-up) CORE A horizontal venturimeter has an inlet area of $0.02\ \mathrm{m^2}$ and a throat area of $0.01\ \mathrm{m^2}$. A mercury–water differential manometer reads 0.1 m and $C_d = 0.96$. Find the discharge of water.

1. $h = x\left(\dfrac{S_h}{S_o} - 1\right) = 0.1 \times (\,13.6 - 1\,) = 1.26\ \mathrm{m\ of\ water}$
2. $\sqrt{2gh} = \sqrt{2 \times 9.81 \times 1.26} = 4.972\ \mathrm{m/s}$
3. $Q_{\mathrm{th}} = \dfrac{0.02 \times 0.01 \times 4.972}{\sqrt{0.02^2 - 0.01^2}} = \dfrac{0.0009944}{0.017321} = 0.05741\ \mathrm{m^3/s}$
4. $Q = C_d\, Q_{\mathrm{th}} = 0.96 \times 0.05741 = 0.0551\ \mathrm{m^3/s}\ (\,55.1\ \mathrm{L/s}\,)$

Example 8.1 HOTS A 300 mm × 150 mm horizontal venturimeter carries oil ($S = 0.9$). A mercury differential manometer reads 20 cm and $C_d = 0.98$. Find the discharge.

1. $h = 0.2 \times \left(\dfrac{13.6}{0.9} - 1\right) = 2.822\ \mathrm{m\ of\ oil}$
2. $a_1 = \dfrac{\pi}{4} \times 0.3^2 = 0.07069\ \mathrm{m^2}$; $a_2 = \dfrac{\pi}{4} \times 0.15^2 = 0.01767\ \mathrm{m^2}$
3. $Q_{\mathrm{th}} = \dfrac{0.07069 \times 0.01767 \times \sqrt{2 \times 9.81 \times 2.822}}{\sqrt{0.07069^2 - 0.01767^2}} = 0.1358\ \mathrm{m^3/s}$
4. $Q = 0.98 \times 0.1358 = 0.1331\ \mathrm{m^3/s}\ (\,133\ \mathrm{L/s}\,)$

Common mistakes and exam tips.

- Using the manometer reading x directly as h. Convert with $x\,(\,S_h/S_o - 1\,)$.
- Using $S_o = 1$ when the pipe carries oil. Use the specific gravity of the flowing liquid.
- Taking areas from diameters in mm. Convert to metres first, then $a = \pi D^2/4$.
- Forgetting C_d. Without it you have only the theoretical discharge.
- Mixing up a_1 and a_2 in the square root. The denominator is $\sqrt{a_1^2 - a_2^2}$ with a_1 the larger area.

Check your understanding.

1. CORE Where is the pressure lowest in a venturimeter? *(At the throat.)*

2. CORE A mercury manometer on a water venturimeter reads 0.1 m. What is h? *($h = 0.1 \times 12.6 = 1.26\,\text{m of water}$.)*
3. HOTS If the manometer reading is increased four times, how does the discharge change? *(It doubles, since $Q \propto \sqrt{h}$.)*

9. Orifice meter (L7) CORE

An orifice meter is a cheaper, more compact substitute for the venturimeter: a thin plate with a sharp-edged circular hole clamped between pipe flanges. It works on the same principle, but the sudden contraction and expansion waste much more energy, so its coefficient of discharge is only about 0.6 to 0.65.

> **IN SIMPLE WORDS**
> An orifice meter is a flat plate with a round hole fitted inside a pipe. The water squeezes through the hole as a jet, and the pressure drops, just as in a venturimeter. It is cheap and takes little space, but the water loses much energy in the swirls after the plate. For this reason its coefficient of discharge is much lower than that of a venturimeter.

9.1 Construction and working CORE

> **IN SIMPLE WORDS**
> After the hole, the jet keeps narrowing for a short distance. The narrowest place is the vena contracta, which is smaller than the hole itself. The pressure is lowest there, so the second pressure tapping is placed at that point. Beyond it the jet widens again into the pipe, with a lot of swirl and energy loss.

1. The orifice diameter is usually about half the pipe diameter.
2. Section 1 (pressure tapping) is about 1.5 to 2 pipe diameters upstream of the plate.
3. The jet leaving the orifice keeps contracting to a minimum area, the **vena contracta** (section 2), a short distance downstream. The second tapping is placed there.
4. Downstream, the jet expands violently to fill the pipe again; most of the pressure difference is lost in the eddies.

9.2 Discharge CORE

> **IN SIMPLE WORDS**
> The orifice-meter formula has the same form as the venturimeter formula. The only change is that the hole area a_0 replaces the throat area. The difference between the jet area and the hole area, and the swirling losses, are all hidden inside the one coefficient C_d, which is about 0.6.

Bernoulli between sections 1 and 2, with the vena contracta area written as $a_2 = C_c \times a_0$ (C_c = coefficient of contraction) and the unknown factors combined into one coefficient, gives the same form as the venturimeter with the orifice area a_0 in place of the throat area:

$$Q = C_d \frac{a_0 a_1 \sqrt{2gh}}{\sqrt{a_1^2 - a_0^2}}$$

Here a_1 is the pipe area, a_0 the orifice area and h the difference of pressure heads, found from the manometer as in Section 8.3.

Derivation from scratch: discharge through an orifice meter

Aim. Prove that $Q = C_d \dfrac{a_0 a_1 \sqrt{2gh}}{\sqrt{a_1^2 - a_0^2}}$.

Assumptions.

1. The flow is steady and incompressible, and ideal between section 1 (pipe, area a_1) and section 2 (vena contracta).
2. The area of the jet at the vena contracta is $a_2 = C_c\, a_0$, where a_0 is the orifice area and C_c is the coefficient of contraction.
3. h is the difference of piezometric heads between sections 1 and 2, found from the manometer as in Section 8.3.

Step 1. Bernoulli and continuity. Exactly as for the venturimeter, with a_2 in place of the throat area.

$$\frac{V_2^2 - V_1^2}{2g} = h, \qquad a_1 V_1 = a_2 V_2$$

Step 2. Solve for V_2.

$$V_2 = \frac{a_1 \sqrt{2gh}}{\sqrt{a_1^2 - a_2^2}} = \frac{a_1 \sqrt{2gh}}{\sqrt{a_1^2 - C_c^2 a_0^2}}$$

Step 3. Theoretical discharge. The jet area is $C_c a_0$.

$$Q_{th} = a_2 V_2 = \frac{C_c\, a_0 a_1 \sqrt{2gh}}{\sqrt{a_1^2 - C_c^2 a_0^2}}$$

Step 4. Allow for losses. Friction makes the real velocity smaller by the coefficient of velocity C_v.

$$Q = C_v\, Q_{th} = \frac{C_c C_v\, a_0 a_1 \sqrt{2gh}}{\sqrt{a_1^2 - C_c^2 a_0^2}}$$

Step 5. Combine the unknown factors into one coefficient. Define C_d so that the formula keeps the venturimeter form with a_0.

$$C_d = C_c C_v \frac{\sqrt{a_1^2 - a_0^2}}{\sqrt{a_1^2 - C_c^2 a_0^2}}$$

Then multiplying C_d by $\dfrac{a_0 a_1 \sqrt{2gh}}{\sqrt{a_1^2 - a_0^2}}$ gives exactly the expression in Step 4.

Result.

$$Q = C_d \frac{a_0 a_1 \sqrt{2gh}}{\sqrt{a_1^2 - a_0^2}}$$

What it tells us. Again $Q \propto \sqrt{h}$. The coefficient C_d (about 0.6 to 0.65) is much smaller than for a venturimeter because the jet contracts ($C_c < 1$) and much energy is lost in the sudden expansion after the plate.

9.3 Venturimeter and orifice meter compared HOTS

Feature	Venturimeter	Orifice meter
Coefficient of discharge	0.95–0.99	0.60–0.65
Permanent pressure loss	Small (gradual divergence)	Large (sudden expansion)
Cost and space	Costly, long	Cheap, fits between flanges
Changing the range	Difficult	Easy: swap the plate
Typical use	Permanent installations, large pipes	Testing, temporary or low-cost metering

Worked example

Step-by-step method.

1. Convert the orifice diameter and the pipe diameter to metres and find a_0 and a_1.
2. Find h from the manometer (Section 8.3), or take it directly if it is given in metres of liquid.
3. Find $\sqrt{2gh}$ and $\sqrt{a_1^2 - a_0^2}$ separately, where a_1 is the pipe area.
4. Compute $Q = C_d\, a_0 a_1 \sqrt{2gh} / \sqrt{a_1^2 - a_0^2}$ using the orifice C_d (about 0.6).
5. State Q in $\mathrm{m^3/s}$ and L/s.

Example 9.0 (warm-up) CORE An orifice meter has an orifice area of $0.005\ \mathrm{m^2}$ in a pipe of area $0.02\ \mathrm{m^2}$. The head difference is 4 m of water and $C_d = 0.6$. Find the discharge.

1. $\sqrt{2gh} = \sqrt{2 \times 9.81 \times 4} = 8.859\ \mathrm{m/s}$
2. $\sqrt{a_1^2 - a_0^2} = \sqrt{0.02^2 - 0.005^2} = 0.019365\ \mathrm{m^2}$
3. $Q = 0.6 \times \dfrac{0.005 \times 0.02 \times 8.859}{0.019365} = 0.6 \times 0.04575 = 0.0274\ \mathrm{m^3/s}\ (27.4\ \mathrm{L/s})$

Example 9.1 CORE An orifice meter with a 100 mm orifice is fitted in a 200 mm water pipe. A mercury–water differential manometer reads 50 cm and $C_d = 0.64$. Find the discharge.

1. $h = 0.5 \times (13.6 - 1) = 6.3\,\text{m of water}$
2. $a_0 = \frac{\pi}{4} \times 0.1^2 = 0.007854\,\text{m}^2$; $a_1 = \frac{\pi}{4} \times 0.2^2 = 0.031416\,\text{m}^2$
3. $Q = \frac{0.64 \times 0.007854 \times 0.031416 \times \sqrt{2 \times 9.81 \times 6.3}}{\sqrt{0.031416^2 - 0.007854^2}} = 0.0577\,\text{m}^3/\text{s}\ (57.7\,\text{L/s})$

Common mistakes and exam tips.

- Using a venturimeter C_d (0.98) for an orifice meter. Use about 0.6.
- Taking the pipe area for a_0. a_0 is the orifice area and a_1 is the pipe area.
- Using the manometer reading as h without converting with $x\,(S_h/S_o - 1)$.
- Putting the tapping at the orifice plate. The second tapping is at the vena contracta.

Check your understanding.

1. CORE Where is the second pressure tapping of an orifice meter placed? *(At the vena contracta.)*
2. CORE What is a typical C_d for an orifice meter? *(About 0.6 to 0.65.)*
3. HOTS Why does an orifice meter waste more energy than a venturimeter? *(The jet expands suddenly after the plate and the eddies destroy the pressure.)*

10. Pitot tube (L8) CORE

A Pitot tube measures the velocity at a point. Its open end faces the flow, so the fluid entering it is brought to rest; the kinetic head is converted into a rise of pressure head, and that rise gives the velocity.

IN SIMPLE WORDS

A Pitot tube is a thin bent tube with its open end pointing into the flow. The moving fluid piles up and stops at the tube's mouth, so all its speed turns into extra pressure there. A pressure gauge or a column of liquid shows this extra pressure, and from it we get the speed at that point. It works like holding your hand out of a moving car, where the push you feel grows with speed.

10.1 Principle CORE

IN SIMPLE WORDS

At the mouth of the tube the fluid is brought to rest, at the stagnation point. Bernoulli's equation says that the speed head that was lost has become a pressure head. So the liquid in the tube rises above the surrounding level by exactly the speed head. Faster flow gives a bigger rise, and the speed is read from the rise h.

Derivation from scratch: velocity from a Pitot tube

Aim. Prove that the velocity at a point is $V_1 = \sqrt{2gh}$, where h is the dynamic head.

Assumptions.

1. The flow is steady, incompressible and frictionless along the streamline from point 1 to the tube mouth.
2. Point 1 is in the undisturbed stream, and point 2 is the stagnation point at the open mouth of the tube. Both are at the same level, so $z_1 = z_2$.
3. The fluid in the tube is at rest, so $V_2 = 0$ at the mouth.

Step 1. Bernoulli's equation between 1 and 2.

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

Step 2. Insert the conditions. The datum terms cancel and $V_2 = 0$.

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} = \frac{p_2}{\rho g}$$

Step 3. Solve for the velocity head.

$$\frac{V_1^2}{2g} = \frac{p_2 - p_1}{\rho g} = h$$

Step 4. Solve for V_1. Multiply by $2g$ and take the square root.

$$V_1^2 = 2gh \quad \Rightarrow \quad V_1 = \sqrt{2gh}$$

Result.

$$V_1 = \sqrt{2gh}, \qquad h = \frac{p_2 - p_1}{\rho g}$$

What it tells us. The rise h of liquid in the tube is the velocity head of the stream, so $V \propto \sqrt{h}$: doubling the speed makes the rise four times larger. In an open channel the pressure at 1 is just the hydrostatic pressure, so h is the height by which the liquid in the tube stands above the free surface.

h is the dynamic head: the rise of liquid in the tube above the static level. The actual velocity is $V = C_v\sqrt{2gh}$, with the coefficient of velocity C_v about 0.98 (often taken as 1).

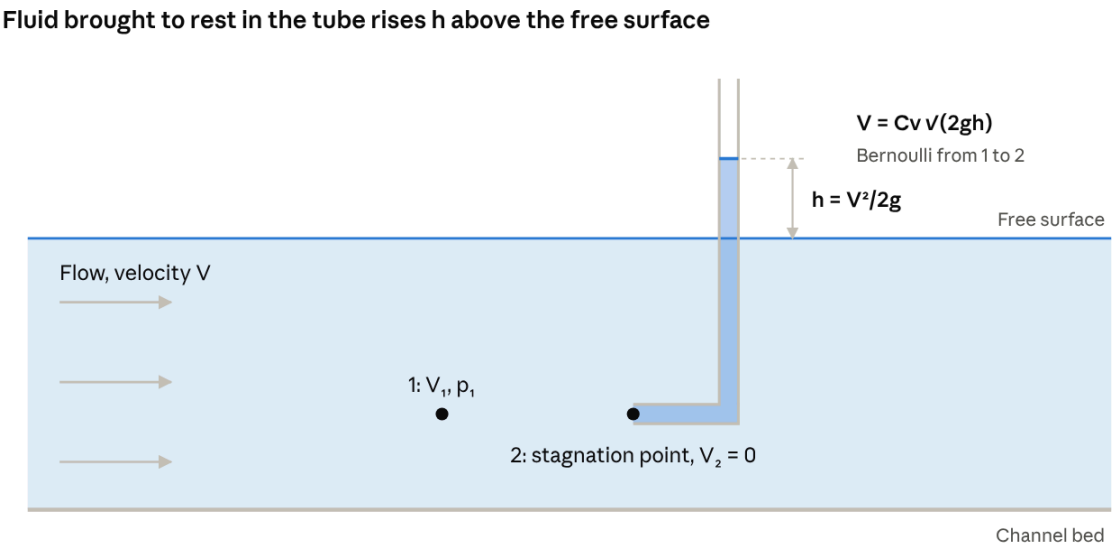


Figure 4. Simple Pitot tube in an open channel

10.2 Arrangements CORE

IN SIMPLE WORDS

There are several ways to read the pressure rise. In a river, a bent glass tube is enough. In a pipe, the pressure is often high, so the Pitot tube is joined to a U-tube manometer. The Pitot-static tube puts both openings in one probe: one faces the flow and one is on the side. It is the instrument used on aircraft to read air speed.

Arrangement	What it reads	Use
Simple Pitot tube (bent glass tube)	Rise h above the free surface	Open channels, rivers
Pitot tube + piezometer in a pipe	Difference between stagnation head and static head	Pipes at low pressure
Pitot tube + differential manometer	$h = x\left(S_h/S_o - 1\right)$	Pipes at any pressure
Pitot-static (Prandtl) tube	Stagnation and static pressure in one probe	Pipes, wind tunnels, aircraft airspeed

The Pitot tube gives the velocity at one point. To find the discharge in a pipe, the velocity is measured at several radii and averaged over the area.

Derivation from scratch: velocity from a Pitot tube with a differential manometer

Aim. Prove that $V = C_v\sqrt{2gx\left(\frac{S_h}{S_o} - 1\right)}$ when the Pitot tube is connected to a U-tube with a heavier liquid.

Assumptions.

1. The Pitot opening (stagnation pressure p_2) and the static opening (pressure p_1) are at the same level in the pipe.
2. The manometer liquid has density ρ_h and the pipe liquid has density ρ_o. The liquids are at rest in the manometer.
3. The reading of the manometer is x, and C_v allows for the small losses.

Step 1. Choose the level. The stagnation pressure is the larger one. Let A–A be the mercury surface in the limb joined to the Pitot opening. Both openings are at height y above A–A. The mercury in the other limb stands higher by x.

Step 2. Pressure at A–A through the Pitot limb.

$$p_{\mathrm{A}} = p_2 + \rho_o g y$$

Step 3. Pressure at A–A through the static limb. There is mercury of height x, then pipe liquid of height $y - x$ up to the static opening.

$$p_{\mathrm{A}} = p_1 + \rho_o g\,(\,y - x\,) + \rho_h g x$$

Step 4. Equate and cancel $\rho_o g y$.

$$p_2 + \rho_o g y = p_1 + \rho_o g y - \rho_o g x + \rho_h g x \quad \Rightarrow \quad p_2 - p_1 = \left(\rho_h - \rho_o\right) g\, x$$

Step 5. Convert to the dynamic head. Divide by $\rho_o g$.

$$h = \frac{p_2 - p_1}{\rho_o g} = x\left(\frac{\rho_h}{\rho_o} - 1\right) = x\left(\frac{S_h}{S_o} - 1\right)$$

Step 6. Use the Pitot-tube result. Put h into $V = C_v\sqrt{2gh}$.

Result.

$$V = C_v\sqrt{2gx\left(\frac{S_h}{S_o} - 1\right)}$$

What it tells us. For water in the pipe and mercury in the manometer, $V = C_v\sqrt{2g \times 12.6\,x}$. The openings are at the same level, so the pipe height does not matter.

Worked examples

Step-by-step method.

1. Find the head h. If a manometer is used, $h = x\,(\,S_h/S_o - 1\,)$ in metres of the pipe liquid.
2. If the head is given in another fluid (for example mm of water for an air flow), convert the pressure difference and use the density of the flowing fluid.
3. Compute $V = C_v\sqrt{2gh}$.
4. If C_v is not given, use 1.
5. State the velocity in m/s (multiply by 3.6 for km/h).

Example 10.0 (warm-up) CORE A Pitot tube in an open channel shows water standing 0.2 m above the free surface. Taking $C_v = 1$, find the velocity of the water at that point.

1. $h = 0.2\ \text{m}$
2. $V = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 0.2} = \sqrt{3.924} = 1.98\ \text{m/s}$

Example 10.1 CORE A Pitot-static tube in a water pipe is connected to a mercury differential manometer that reads 100 mm. Taking $C_v = 0.98$, find the velocity at that point.

1. $h = 0.1 \times (\,13.6 - 1\,) = 1.26\ \text{m of water}$
2. $V = 0.98 \times \sqrt{2 \times 9.81 \times 1.26} = 4.87\ \text{m/s}$

Example 10.2 HOTS A Pitot-static tube on an aircraft model in a wind tunnel shows a pressure difference of 50 mm of water. Air density is 1.2 kg/m³; take $C_v = 1$. Find the air speed.

1. $p_2 - p_1 = 1000 \times 9.81 \times 0.05 = 490.5\ \text{N/m}^2$
2. $V = \sqrt{\dfrac{2 \times 490.5}{1.2}} = 28.6\ \text{m/s}\ (\,103\ \text{km/h}\,)$
3. Note the head is converted to metres of **air**: $h = \dfrac{490.5}{1.2 \times 9.81} = 41.7\ \text{m}$

Common mistakes and exam tips.

- Using the head of water for an air flow. Convert the pressure to metres of air: $h = \Delta p / \left(\,\rho_{\text{air}} g\,\right)$, or use $V = \sqrt{2\Delta p/\rho}$.
- Forgetting that a Pitot tube gives the velocity at one point only, not the mean velocity in the pipe.
- Using the manometer reading as h without the factor $(\,S_h/S_o - 1\,)$.
- Dropping C_v when it is given. The actual velocity is $C_v\sqrt{2gh}$.

Check your understanding.

1. CORE What is the velocity at the stagnation point? *(Zero.)*
2. CORE A Pitot tube shows $h = 0.8\ \text{m}$ ($C_v = 1$). Find V. *($\sqrt{2 \times 9.81 \times 0.8} = 3.96\ \text{m/s}$.)*

3. HOTS The flow speed doubles. How does the rise h change? *(It becomes four times as large, since $h \propto V^2$.)*

11. Formula sheet and practice questions CORE

11.1 Formula sheet CORE

Topic	Result
Newton's law of viscosity	$\tau = \mu\, du/dy$; thin film $\tau = \mu U/t$
Kinematic viscosity	$\nu = \mu/\rho$; 1 stoke $= 10^{-4}$ m^2/s; 1 poise $= 0.1$ N$\cdot$s/m^2
Mass properties	$w = \rho g$; $v = 1/\rho$; $S = \rho/\rho_{\text{water}}$
Journal bearing	$U = \pi DN/60$; $T = \mu U/t \times \pi DL \times D/2$; $P = 2\pi NT/60$
Bulk modulus	$K = -dp/(dV/V) = dp/(d\rho/\rho)$; gas: $K = p$ (isothermal), kp (adiabatic)
Surface tension	Droplet $4\sigma/d$; bubble $8\sigma/d$; jet $2\sigma/d$
Capillarity	$h = 4\sigma\cos\theta/(\rho g d)$
Euler's equation	$dp/\rho + g\,dz + v\,dv = 0$
Bernoulli's equation	$p/\rho g + V^2/2g + z =$ constant; real fluid: $+ h_L$
Venturimeter	$Q = C_d\, a_1 a_2 \sqrt{2gh}\,/\sqrt{a_1^2 - a_2^2}$, C_d 0.95–0.99
Orifice meter	$Q = C_d\, a_0 a_1 \sqrt{2gh}\,/\sqrt{a_1^2 - a_0^2}$, C_d 0.60–0.65
Manometer head	$h = x\,(S_h/S_o - 1)$; $h = x\,(1 - S_l/S_o)$
Pitot tube	$V = C_v\sqrt{2gh}$

11.2 Short-answer questions (2 marks) CORE

1. CORE Define a fluid. How does it differ from a solid under shear?

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

2. ADVANCED What is the continuum hypothesis? When does it fail?

3. CORE State Newton's law of viscosity and define a Newtonian fluid.
4. CORE Distinguish between dynamic and kinematic viscosity, with their SI units and dimensions.
5. HOTS Why does the viscosity of a liquid fall, and that of a gas rise, with temperature?
6. CORE Define bulk modulus and compressibility.
7. CORE Show that the pressure inside a soap bubble is $8\sigma/d$.
8. HOTS Why does water rise and mercury fall in a capillary tube?

9. CORE State Bernoulli's equation and its assumptions.
10. CORE What is the vena contracta?
11. HOTS Why is the divergent cone of a venturimeter longer than the convergent cone?
12. HOTS Distinguish a Pitot tube from a Pitot-static tube.

11.3 Long-answer and numerical questions (10 marks) CORE

1. CORE Classify fluids by their rheological behaviour, with a shear stress–velocity gradient diagram and examples.
2. CORE Derive Euler's equation of motion along a streamline and integrate it to obtain Bernoulli's equation.
3. CORE Derive the expression for discharge through a venturimeter. How is h found from a differential manometer?
4. HOTS Derive the expression for discharge through an orifice meter and compare it with the venturimeter.
5. CORE Derive the expression for capillary rise and the pressure inside a droplet, a bubble and a jet.
6. CORE Explain the principle of the Pitot tube and its use for measuring velocity in a pipe.
7. CORE Glycerine has a kinematic viscosity of 11.8 stokes and a specific gravity of 1.26. Find its dynamic viscosity. *(Ans:* $1.49\ \mathrm{N\cdot s/m^2}$*)*
8. CORE 1.5 m³ of a liquid weighs 12 kN. Find its specific weight, density and specific gravity. *(Ans:* $8000\ \mathrm{N/m^3}$, $815.5\ \mathrm{kg/m^3}$, 0.815*)*
9. CORE An 80 mm shaft rotates at 1200 rpm in a 100 mm long bearing with an oil film 0.2 mm thick ($\mu = 0.08\ \mathrm{N\cdot s/m^2}$). Find the torque and power lost. *(Ans:* $2.02\ \mathrm{N\cdot m}$, $254\ \mathrm{W}$ *)*
10. CORE Find the pressure increase needed to reduce the volume of water by 1% ($K = 2.2\times10^9\ \mathrm{N/m^2}$). *(Ans:* $22\ \mathrm{MN/m^2}$*)*
11. CORE Find the capillary rise of water in a 3 mm glass tube ($\sigma = 0.0725\ \mathrm{N/m}$, $\theta = 0$). *(Ans:* $9.85\ \mathrm{mm}$*)*
12. HOTS Water flows horizontally from a 100 mm pipe into a 50 mm pipe at 0.01 m³/s. The pressure in the larger pipe is 200 kN/m². Neglecting losses, find the pressure in the smaller pipe. *(Ans:* $187.8\ \mathrm{kN/m^2}$*)*
13. CORE A 200 mm × 100 mm horizontal venturimeter carries water. A mercury manometer reads 25 cm and $C_d = 0.97$. Find the discharge. *(Ans:* $0.0619\ \mathrm{m^3/s}$*)*
14. CORE An orifice meter with a 150 mm orifice in a 300 mm pipe shows a pressure difference of 4 m of water. Taking $C_d = 0.62$, find the discharge. *(Ans:* $0.100\ \mathrm{m^3/s}$*)*
15. CORE A Pitot tube in a water pipe shows a dynamic head of 0.3 m. Taking $C_v = 0.98$, find the velocity. *(Ans:* $2.38\ \mathrm{m/s}$*)*